



FEDERAL MINISTRY OF EDUCATION
SENIOR SECONDARY EDUCATION CURRICULUM



BIOLOGY

FOR
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**NIGERIAN EDUCATIONAL RESEARCH AND
DEVELOPMENT COUNCIL (NERDC)**



**BIOLOGY
FOR
SENIOR SEONDARY SCHOOL**

SENIOR SECONDARY SCHOOL 1-3

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**NIGERIAN EDUCATION RESEARCH AND
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FOREWORD

The Nigerian Educational Research and Development Council (NERDC) has the mandate to develop curricula for use at all levels of the educational system in Nigeria. In line with government reform in education the NERDC was directed by the National Council on Education (NCE) to re-view and re-align the existing senior secondary school curricula to meet the targets of the reform in the context of National Economic Empowerment and Development Strategies (NEEDS) and the Millennium Development Goals (MDG). I am pleased to note that the NERDC has developed the Senior Secondary School Curriculum.

It is therefore for me, a great privilege to present the curricula to all Nigerians for the use of our children now and in the future. I must congratulate all those who contributed to the development of these curricula. I wish to thank most especially the Nigerian Educational Research and Development Council for keeping faith with its mandate, the High Level Policy Committee on Curriculum Development for preparing the policy grounds for the curricula, and the resource person for a job well done.

It is my fervent hope that the teachers and learners for whom these curricula are produced would demonstrate commitment and assiduity in using these curricula. This is a proud legacy to leave for posterity.

With great expectations, I gladly recommend these curricula to all for the purpose of producing the best textual materials, the best in teaching performance and the best learning outcome, and most importantly, for attaining the goals we have set for ourselves in education in line with the Millennium Development Goals (MDGS) and in compliance with the National Economic Empowerment and Development Strategies (NEEDS).

DR. Sam Omiyi Egbu
Honourable Minister of Education,
Abuja.

PREFACE

Following the Federal Government reform in education and the need to attain the Millennium Development Goals (MDGs) and the critical targets of the National Economic Empowerment and Development Strategies (NEEDS), which can be summarized as: value- reorientation, poverty eradication, job creation, wealth generation and using education to empower the people, it has become imperative that the existing curricula for senior secondary school curricula should be reviewed and re-aligned to fit the reform programme. The National Council on Education (NCE) at its meeting in Ibadan in December 2005, directed the NERDC to carry out this assignment. The NCE also approved a new Senior Secondary School curriculum structure namely: Senior Secondary School (Science and Mathematics), Senior Secondary School (Humanities), Senior Secondary School (Business) and Senior Secondary School (Technology), listing relevant subjects for each Stream.

In response to these developments, a High Level Policy Committee on Curriculum Development (HLPC), made up of critical stakeholders and chaired by NERDC, took the initiative to provide the guidelines for re-structuring the curriculum.

Between January 2007 and March 2008, the NERDC convened a meeting of experts and also organized several workshops to produce the Senior Secondary School Curriculum, which would ensure continuity and flow of themes, topics and experiences from Senior Secondary one to Senior Secondary School junior secondary three levels.

The Curriculum reflects depth, appropriateness, and interrelatedness of the curricula contents. Also, emerging issues which covered value orientation, peace and dialogue, including human rights education, family life/HIV and AIDS education, entrepreneurial skills et cetera were infused into the relevant contents of the new Senior Secondary School Curriculum.

In general, the curriculum pays particular attention to the achievement of the Millennium Development Goals (MDGs) and the critical elements of the National Economic Empowerment and Development Strategies (NEEDS).

Since the curriculum represents the total experiences to which all learners must be exposed, the contents, performance objectives, activities for both teachers and learners, teaching and learning materials and evaluation guide are provided. The prescriptions represent the minimum content to be taught in the schools in order to achieve the objectives of the new Senior Secondary School programme. However, teachers are encouraged to enrich the contents with relevant materials and information from their immediate environment, but adapting the curriculum to their needs and aspirations. Thus the curriculum can be adapted for such special needs as nomadic education, non-formal education and education of the physically challenged.

In conclusion, I wish to express profound gratitude to the President of the Federal Republic of Nigeria, Umaru Musa Yar'Adua, for his vision and tenacity and his financial support for the review and effective implementation of the Senior Secondary School Curriculum at all levels; all of which came through the funds allocated to the attainment of the MDGs.

I acknowledge the efforts of the Curriculum Development Centre of the NERDC which coordinated the various activities and workshops that led to the development of the Senior Secondary School Curriculum. The planning, writing and critique teams which were drawn from all parts of this country also deserve commendation.

It is our belief that when these curricula are properly implemented, the future generations of this great country will be better for it.

Prof. Godswill Obioma fman, fiica, fcon, fnae
Executive Secretary
NERDC.

SENIOR SECONDARY SCHOOL CURRICULUM

BIOLOGY

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BIOLOGY CURRICULUM FOR SENIOR SECONDARY SCHOOLS

INTRODUCTION:

This Biology Curriculum was adapted and revised from 1985 edition developed by Comparative Education Study and Adaptation Centre (CESAC). The objectives of this curriculum have been derived from the National Policy on Education (2004) and the main objectives are to prepare students to acquire:

1. Adequate laboratory and field skills in biology;
2. Meaningful and relevant knowledge in biology;
3. Ability to apply scientific knowledge to everyday life in matters of personal and community health and agriculture;
4. Reasonable and functional scientific attitude.

In pursuance of the stated objectives, the contents and context of the curriculum places emphasizes on field studies, guided discovery, laboratory techniques and skills along with conceptual thinking.

The curriculum intended to provide a modern biology course as well as meet the needs of the learner and the society through relevant and functionality in its contents, methods, processes and applications.

It covers the major themes of:

- Theme 1: Organization of Life
- 2: Organisms at Work
- 3: The Organisms and its Environment
- 4: Continuity of Life

These themes are of direct relevance to the society and the learner.

In planning the new biology curriculum, the spiral approach to sequencing a science course was adopted. In the approach, the concepts to be taught are arranged in such a way that they run throughout the three-year post basic course, with the concepts being discussed in greater depth as the course progresses.

The curriculum is organized into six sections:

1. Topic
2. Performance Objectives
3. Content
4. Activities – Teacher and Students
5. Teaching and Learning Materials
6. Evaluation Guide

Such organisation provides maximum guide to the teacher. Teachers should, however, note that the performance objectives presented in this column are not exhaustive as the need arises, especially in the higher cognitive level and psychomotor and affective domains. This curriculum provides a logical and psychological follow up for the 9-Year Basic Science Core Curriculum.

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THEME 1: ORGANISATION OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
1. RECOGNIZING LIVING THINGS	Students should be able to: 1. State the characteristics of living things 2. Give examples of levels of organization of life 3. State complexity of organization in higher organisms	1. Characteristics of living things 2. Differences between plants and animals 3. Organisation of life (a) levels of organisation of life: - Cell (Euglena, paramecium) - Tissue (hydra) - Organ (Onion bulb, the heart of a cow) - System (e.g. digestive system, excretory system) (b) Complexity of organisation in higher organisms: advantages and disadvantages	1. Teacher provides a live cockroach and a potted plant. Teacher asks students to observe. 2. Teacher prepares or brings prepared microscope slides of Euglena or Paramecium, and Hydra. Mounts them on the microscope. 3. Teacher provides onion bulbs, cow's heart and the digestive system of a goat.	1. Students collect as many living things as possible from the school premises and record them. 2. Students view the slides under the microscope and record their observations. 3. Students observe the specimens, record their observations and make inferences.	1. Plants and animals in the environment 2. Slides of unicellular organisms, onion bulbs, cow's heart, alimentary canal of a goat 3. Charts of the above organisms 4. Models of heart and digestive system of some animals	Students to: 1. List three characteristics of living things 2. State three differences between plants and animals 3. State two advantages and disadvantages each of complexity in higher organisms

THEME 1: ORGANISATION OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
2. CLASSIFICATION OF LIVING THINGS (1) KINGDOMS MONERA, PROTISTA AND FUNGI, PLANTAE AND ANIMALIA	Students should be able to: 1. State the characteristic features of the kingdoms and give specific examples of representative organisms.	1. Kingdom Monera (Prokaryotes): single-celled, motile or non-motile organisms. No definite nucleus. Bacteria and blue-green algae make up this kingdom. 2. Kingdom Protista (Eukaryotes) : Single-celled, motile or non-motile organisms. Complex Cell structure with definite nucleus e.g. Chlamydomonas and Amoeba 3. Kingdom Fungi (Eukaryotes): mainly non-motile organisms, composed of hyphae containing nuclei e.g. moulds, mushrooms and Rhizopus 4. Kingdom Plantae (Eucaryotes) Many-celled, non-motile organisms which contain chlorophyll that enable them to photosynthesize. Members include mosses, ferns, pines, oil palms and yam plants	1. Teacher to grow cultures of bacteria and blue-green algae 2. Teacher prepares slides of Chlamydomonas or Amoeba 3. Teacher guides the students to collect ditch water and view samples under the microscope. 4. Teacher guides the students to uproot the fern plant, collect lemna, pistia (water lettuce) and breaks some branches of shrubs and brings them into the laboratory for students' observation	1. Students to observe the cultures, record and make inference 2. Students view the slides and ditch water under the microscope, record their observations and make inferences 3. Students observe and record observations 4. Students observe, draw and make inferences 5. Students observe all the specimens, record and make inferences	Cultures of bacteria and blue-green algae Microscope, microscope slide, cover slips and ditch water. Prepare slides of Amoeba, Chlamydomonas etc. Mushrooms or moulds Moss plant, lemna, pistia, flowering plants Charts/drawings of all the organisms listed above	Students to: 1. State how to grow Rhizopus 2. Students should be able to describe single celled organisms 3. Describe Rhizopus and mushrooms 4. Describe structures habitats and adaptive features of moss, fern, lemna, pistia and flowering plants

SS 1

THEME 1: ORGANISATION OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
		5. Kingdom Animalia (Eucaryotes) Many-celled, motile organisms that feed on other organisms. Members include corals, worms, insects, snails, fishes, frogs, snakes, monkeys and cows	5. Teacher guides students to collect earthworms, cockroaches, grasshoppers, snails, frogs or toads and bring them into the laboratory for the students to observe and record		Snails, toads, cockroaches, grasshoppers, earthworms etc Pictures, drawings and charts of all the organisms	5. Describe structures, habitats and habitat adaptive features of earthworm, cockroach, snail, toad etc

SS 1

THEME 1: ORGANISATION OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
3. THE CELL	The students should be able to: 1. Recognise single celled organisms 2. Distinguish between free-living cells and colonies, filaments and tissues 3. Describe the general structure of a cell 4. Differentiate between a plant and an animal cell	1. Cell as a living unit of an organism 2. Forms in which living cells exist: i. independent organism ii. as a colony iii. as a filament 3. Cell structure i. the cell theory ii. cell structure and functions of cell components iii. differences and similarities between plant and animal cells	1. Teacher provides prepared slides of Paramecium or Euglena, Volvox and Spirogyra. 2. Teacher guides students to collect Spirogyra from pond or ditch, feel, observe and record their observations 3. Teacher mount slides of plant and animal cells for students' observation 4 Teacher presents charts showing similarities and differences in plant and animal cells	1. Students observe the slides under the microscope and record their observations 2. Students observe the spirogyra and note its nature 3. Students observe slides and charts of plant and animal cells draw and note their differences and similarities	• Microscope • Slides • Prepared slides • Charts showing different forms in which cells exist • A generalized diagram of a cell • Diagrams of plant and animal cells	Students to: 1. State why the cell is a living unit 2. Identify the various forms in which cells exist. 3. Describe the structure of the cell 4. Differentiate /distinguish a plant cell from an animal cell.

THEME 1: ORGANISATION OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
4. THE CELL AND ITS ENVIRONMENT	Students should be able to: 1. Demonstrate diffusion and osmosis experimentally 2. Recognise that osmosis is a form of diffusion 3. Recognise that plasmolysis can lead to wilting and haemolysis can lead to loss of blood	1. Diffusion: i. definition ii. process iii. significance 2. Osmosis i. diffusion of water through a selectively permeable membrane ii. haemolysis iii. plasmolysis iv. osmometer with living material v. biological significance of these processes	1. Teacher provides cheap perfume and requests a student to spray it one end of the classroom. 2. The teacher places a crystal of potassium permanganate in a beaker of water and leave it to stand and ask the students to observe what has happened 3. Teacher to demonstrate osmosis in a (i) a non-living system and (ii) a living system 4. Teacher demonstrates experiments to show effects of solutions of different concentrations on red blood cells; effects of water and sugar solution on onion cells	1. Student sprays the perfume as directed by the teacher. 2. Students describe what happened 3. Students repeat the experiments, observe, record and discuss their observations	Perfume, potassium permanganate, beakers and water Pig's bladder, thistle funnel, sugar, peeled yam stopper and clamp	Students to: 1. State definitions of the processes. 2. Carry out experiments to demonstrate the processes

THEME 1: ORGANISATION OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
5. SOME PROPERTIES AND FUNCTIONS OF THE CELL	Students should be able to: 1. Recognise that some nutrients (micronutrients) are needed in small quantities while others (macronutrients) are needed in large quantities 2. Show experimentally that the break-down of carbohydrates may be partial (fermentation) or complete 3. Recognise that cells require proteins, fats and carbohydrates for the production of new protoplasm, for repair, growth and provision of energy	1. Feeding definition and types (i) micronutrients (ii) macronutrients 2. Cellular respiration (i) definition of cellular respiration (ii) aerobic respiration (catabolism); Krebs cycle (iii) anaerobic respiration (iv) energy release during respiration 3. Anabolism – usefulness of food 4. Autotrophy (i) photosynthesis (ii) chemosynthesis	1. Teacher sets up experiments to show: (a) the effects of different nutrients on Spirogyra (b) the effects of high concentration of fertilizer on plants 2. Teacher sets experiments to show: (a) respiration in yeast (anaerobic) (b) respiration in rat (aerobic) 3. Teacher draws the Krebs cycle on the chalkboard 4. Teacher sets experiments to show growth as: (a) increase in gross height (b) increase in length	1. Students to observe, record and discuss experiments 2. Students draw the Krebs cycle along with the teacher 3. Students produce saliva from their mouths to carry out the experiments 4. Students observe the experiments and to identify the gas produced using a glowing splint	Spirogyra, fertilizer, Yeast, and Rat Conical flask, boiling tube, Elodea plant Sodium hydroxide rubber bung, conical flask, potted plants, clamps and girdles Vaseline Maize grains, cowpea seeds, empty milk tins or plastic containers	Students to: 1. State different stages of the Krebs Cycle 2. Distinguish between aerobic and anaerobic respiration 3. Discuss the uses of food to organisms 4. Differentiate between autotrophs and heterotrophs 5. State three basic materials necessary for photosynthesis
	4. Recognise that certain cells are autotrophic and others are heterotrophic 5. Discuss the role of enzymes in digestion	5. Heterotrophy 6. Role of enzymes 7. Excretion 8. Excretory organelles in living cells	5. Teacher helps students to prepare and mount slides of root tip to observe cell division 6. Teacher guide students to demonstrate action of ptyalin on cooked starch	5. Students observe the experiments, record and discuss their observations	Auxanometer to record shoot growth, Clinostat, seedlings of cowpea or maize	

THEME 1: ORGANISATION OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
	6. Infer that excretion is the removal of metabolic waste products from the cell which may be toxic or which are in excess of the cell needs. removal of waste products is by diffusion through the body contractile vacuole. 7. Identify growth as (i) increase in dry weight (ii) irreversible increase in size and length (iii) increase in number of cells 8. Carry out experiments (i) from which brings growth curves could be constructed (ii) determine the regions of fastest growth e.g. in the root. 9. Demonstrate by experiments factors that affect growth. 10. Explain ability of cell to detect and respond to external stimuli.	(ii) waste products of metabolic activities of living cells (iii) forms in which waste products are excreted 8. Growth: (i) basis of growth-cell division (mitosis) enlargement and differentiation (ii) regulation of growth by hormones 9. Cell reactions to its environment (i) Irritability as basic characteristics of the protoplasm (iii) Types of responses 10. Movement: (i) Cyclosis (ii) Organelles for movement (iii) Growth movement as regulated by auxins	7. Teacher sets up experiments to demonstrate the release of oxygen during photosynthesis 8. Teacher performs experiment to show (a) that carbon dioxide is used by green plants in sunlight (b) that starch is formed in photosynthesis 9. Teacher performs experiments to show factors that affect plant and animal growth such as availability of nutrients, water, humidity, light, temperature and hormones. 10. Teacher performs experiments to show (a) phototropic response of shoots (b) geotropic response of roots (c) geotropic response of shoots (d) phototactic response in earthworms 11. Teacher makes (a) cultures of Paramecium and Euglena and mount them under the microscope (b) Teacher provides prepared slides of Paramecium and Euglena	6. Students to plant maize grains and cowpea seeds and watch them germinate and grow. Taking note of the length of stem and number of leaves produced and record daily 7. Students observe experiments, record and draw up conclusions 8. Students observe, record, and summarize results in a tabular form 9. Students observe movement of Paramecium and Euglena. Students observe cilia and flagella using prepared slides	Prepared slides of Paramecium and Euglena. Microscope, cultures of Paramecium and Euglena Potted plants, cut out shoots of plants, earthworm	6. State and discuss the role of enzymes in digestion 7. Define excretion in relation to cells 8. Define growth in relation to cell division 9. Describe the factors that affect growth 10. Discuss the cell's ability to respond to stimuli 11. Define reproduction and state the forms of reproduction 12. Differentiate between mitosis and meiosis.

THEME 1: ORGANISATION OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
	11. Identify structural adaptations for mobility e.g. cilia and flagella. 12. Recognise reproduction as the ability of living things to produce new individuals of their type	11. Reproduction: Forms of reproduction a) Asexual: (i) Fission. (ii) Budding. (iii) Vegetative propagation. b) Sexual: (i) Conjugation. (ii) Fusion of male and female gametes, meiosis, structure and functions of the male and female gonads.	12. (a) Teacher provides yeast, warm water and plastic bowls (b) teacher provides prepared slides of conjugation in Paramecium. 13. (a) Teacher provides prepared slides of conjugation in Spirogyra and longitudinal section of the pistil of a flower. (b) Teacher gets students to collect flamboyant flower.	10. (a) Students make temporary mount of yeast cells, Identify and draw yeast cells that show budding. (b) Students examine mounted Paramecium, Identify and draw conjugating Paramecia. 11. Students to observe, record and discuss with the teacher taking note of ovule which contains the ovum 12. Students examine the flower, taking note of all the parts especially the stamen and pistil	- Yeast, warm water, plastic bowls - Flowers of different plants Flamboyant flowers, razor blade	

THEME 2: THE ORGANISM AT WORK

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
6. TISSUES AND SUPPORTING SYSTEMS	Students should be able to: 1. Recognise different skeletal and supporting tissues. 2. State the location and arrangement of skeletal and supporting tissues in animals. 3. State the different supporting tissues (turgid parenchyma, collenchyma, xylem (wood) sclerenchyma) in plants, and the arrangement of these supporting tissues. 4. State the functions of skeleton and supporting tissues in animals and plants and how these functions are performed.	1. Skeleton and supporting systems in animals: (i) Biological significance. (ii) Forms e.g. chitin, cartilages and bones. 2. Types of skeleton: (i) Hydrostatic skeleton. (ii) Exo-skeleton. (iii) Endo-skeleton. 3. Vertebrate skeletons: - the skull, - sternum and ribs. - limb girdles. - limbs. 4. Bones of the vertebral column. 5. Different types of supporting tissues in plants: – location and arrangement of structural components.	1. Teacher provides: cockroaches, grasshoppers, centipede, earthworm and cartilaginous fish, individual bones of the mammalian skeleton and infant mammalian skeleton. 2. Teacher provides transverse section of stems and roots of monocotyledonous and dicotyledonous plants. 3. Teacher to provide relevant materials for students to make working model of a human arm. 4. Teacher to provide small mammal and guide students to produce a mammalian skeleton.	1. Students observe all the specimens taking note of the different skeletons and stating the types of skeleton. 2. Students to identify and list the main parts of the mammalian skeleton. 3. Students view transverse section of roots and stems under the microscope and make drawings of the tissues, taking note of the shape of the cells of the various tissues. 4. Students to list differences between monocotyledonous and dicotyledonous plants. 5. Students to observe the protective functions of the shell of snails, skull of toad. 6. Students to observe the bones and attached muscles in the hind limb of a toad.	Arthropods, earthworms, cartilaginous fish, bones of animals, mammalian skeleton, Monocotyledonous and dicotyledonous plants e.g. guinea grass and waterleaf; snails and toads.	Students to: 1. State three differences between Monocotyledonous and dicotyledonous plant. 2. State the relationship between skeleton and muscles during movement 3. Discuss what plants and animals would look like without exo- and endo-skeletons. 4. Use the boiling or burying method to produce the skeleton of a small mammal.

THEME 2: THE ORGANISM AT WORK

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
	5. Construct a working model of a human arm and explain how it functions	6. Functions of skeleton in animals: (i) protection (ii) support (iii) locomotion 7. Functions of supporting tissues in plants: (i) strength (ii) rigidity (iii) resistance against the forces of wind and water	5. Teacher to provide different bones of the vertebral column and guide the students to identify their features.	7. Students to extract skeleton using small mammal either by boiling or burying method. 8. Students to note the distinguishing features of the bones of the vertebral column.	- Small mammal - Pot - Heating source e.g. stove and cooker. - Methylated spirit, bleach - Different bone specimens from the vertebral column.	5. Identify the characteristics of different bones of the vertebral column

THEME 2: THE ORGANISM AT WORK

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
7. NUTRITION IN ANIMALS	Students should be able to: 1. Define what food substances are. 2. List types of food substances and state what makes a balanced diet is. 3. List types of heterotrophic nutrition and feeding mechanisms in holozoic organisms. 4. (i) List types of mammalian teeth. (ii) Describe the structure of a mammalian tooth. (iii) State the dental formula and adaptations of dentition to mode of nutrition. 5. Describe digestion in Amoeba, Earthworm, Cockroach, grasshopper and birds.	1. Food Substances. (a) Types of food substances. (i) Carbohydrates. (ii) Fats and Oils. (iii) Proteins. (iv) Mineral salts. (v) Vitamins. (vi) Water. (b) Balanced diet. (c) Heterotrophic Nutrition: (i) Types of heterotrophic nutrition. - Holozoic nutrition. - Saprophytic nutrition - Parasitic nutrition.	1. Teacher provides garri, yam, rice, meat, butter, olive oil, common salt, fish, prawn, pepper, cowpea, crabs and onion, etc to the class. 2. Teacher performs experiments to (a) show emulsification of fats or oils (b) Identification of different food types. 3. Teacher performs experiment to show that yeast cells produce a digestive enzyme that degrade food. 4. The teacher exposes some pieces of meat in the laboratory. 5. Teacher collects different types of teeth from the Abattoir and from killed cats or dogs.	1. Students categorise the food items under four classes of food. 2. Students observe the experiments and make inferences. 3. The students grow Rhizopus on bread and observe the degradation of the bread daily. 4. Students observe flies feed on them and record their observations. 5. Students observe the different teeth, draw and label.	1. Many types of food. 2. Charts showing the constituents of a balanced diet. 3. Olive oil, bile from bird and yeast, fresh meat 4 Materials for performing food tests e.g. Fehling's solution and Benedict's solution. Teeth of herbivores and carnivores. Chart/diagram illustrating the internal structure of a mammalian tooth	Students to: 1. List five food items that contains fats or oils, protein and carbohydrate. 2. Describe a meal that is balanced and identify the food it contains substances. 3. Distinguish between heterotrophic and holozoic feeding. 4. List the various heterotrophic nutrition. 5. List types of teeth. 6. Draw and label a mammalian teeth.

THEME 2: THE ORGANISM AT WORK

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
		(ii) Feeding mechanisms in holozoid organisms: - Filter feeding - Deposit feeding - Fluid feeding 2. Mammalian teeth: (a) The different types of teeth (i) Incisors. (ii) Canines. (iii) Premolars. (iv) Molars. (b) Structure of a tooth. (c) Dental formula . (d) Adaptations of dentition to mode of nutrition. 3. Enzymes: (a) what are enzymes, (b) characteristics of enzymes (c) classification of enzymes (d) importance of enzymes	6. Teacher to dissect a cockroach and a pigeon to show the alimentary canals. 7. The teacher performs experiment to test for the acidity of the enzyme ptyalin.	6. Students observe, draw and label. 7. Students observe the experiment, record and discuss.	Pigeon and cockroach. Saliva, test tubes, Benedict's solution, starch, iodine solution, distilled water.	Students to state the importance of the crop in the bird's alimentary canal. State the functions of ptyalin in food digestion. List the various types of enzymes and state their importance in digestion.

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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8. BASIC ECOLOGICAL CONCEPTS	Students should be able to: 1. Name components of any ecosystem. 2. Measure or estimate sizes of some ecosystems. 3. Relate the dynamic nature of an ecosystem to its size. 4. Mention major local (Nigerian) biotic communities. 5. Give names of organisms typical of each community. 6. Briefly describe different types of communities in (a) tropical regions (b) temperate regions. 7. Name ecological factors common to all habitats. 8. Briefly describe each of the factors pointing out their relative importance. 9. Mention the factors which affect water-retentivity of soil, types and determine the amount of water each soil type can hold.	Ecosystem – Components and Sizes: (i) Ecological concepts - environment - biosphere - lithosphere - hydrosphere - atmosphere - habitats - niche - population - biotic community (biome) - ecosystem (ii) Components of an ecosystem - abiotic, biotic, aquatic, terrestrial	1. Teacher takes students on a fieldtrip to at least one of the biotic communities e.g. a forest reserve or a botanic garden. 2. Teacher makes available study charts, photographs, and films of different biotic communities in Nigeria and biomes of the world. 3. The teacher guides the students to study what happens to a cleared piece of land and also to study the effects of food shortage using mice. 4. Teacher guides the students to measure sizes of the ecosystem. 5. Teacher displays ecological instruments.	1. Students study the community, record and discuss. 2. Students study them and discuss. 3. Students to observe the rats, discuss and make conclusions. 4. Students with the help of the teacher perform the experiment to compare water holding capacity in the three samples of soil.	Charts, photographs, films. Mice food. Loamy soil, sandy soil, clay soil, measuring cylinder, cotton wool, funnel and water.	Student to: 1. Define ecosystem, niche and habitat. 2. List and discuss the major biotic communities in Nigeria. 3. State the differences between desert and sahel savanna. 4. State the characteristics of different soil types and link the characteristics to the water retaining capacity of the soil.

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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		2. Local biotic communities or biomes. (i) Tropical rainforest. (ii) Southern guinea savanna. (iii) Northern guinea savanna. (iv) Sahel Savanna. (v) Desert. (vi) Swamp/estuarine regions. 3. Major biomes of the world: (i) Tropical forest (ii) Savanna (iii) Desert (iv) Shrub (v) Afro-alpine (vi) Swamp 4. Population studies by sampling method: (i) population size (ii) dominance (iii) density (iv) factors that affect population.	6. Teacher displays the ecological instruments before the students and make them to use them to measure ecological factors. 7. Teacher takes students to the field and show them how to measure the population of a small area. 8. Teacher provides loamy soil, sandy soil and clay soil, measuring cylinder, cotton wool, funnel and water. 9. show students how to use various measuring instruments to measure the physical factors of the ecosystems e.g. temperature, rainfall etc.	5. Students to study one or more of these ecosystems: a rotten log, a puddle and a pond, a stream and river, a bush. 6. Students study the instruments and write their uses. 7. Students use the materials provided to measure the population of a small area. 8. Students make histograms, bar charts and graph to show different species in a community.	Dry and wet thermometer, soil thermometer, rain gauge, quadrat, fish trap, a pooter, insect net, specimen, bottles etc.	5. Describe the major biomes of the world and give the characteristics of each biome. 6. List the various factors affecting the population of an area. 7. Use the aid of a histogram, show the various animal/plant species in a given area. 8. List some measuring instruments and their uses.

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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		5. Ecological factor: aquatic, terrestrial, and factors common to all habitats. Importance of ecological factors to populations of animals and plants. 6. Relationship between soil types and water holding effect of soil on vegetation. 7. Simple measurement of ecological factors. (i) (a) physical factors: climatic, topographic and gaseous (b) edaphic factors: chemical and physical composition, moisture content and thickness of soil. (ii) Measuring instruments.	10. Show students how to improvise some measuring instruments e.g. wind vane.	9. Students measure some of the ecological factors which they have studied e.g. temperature.	Different types of measuring instruments e.g. thermometer, rain gauge, meter-rule, measuring cylinder, wind-vane.	

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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9. FUNCTIONING ECOSYSTEM: I. Autotrophy and Heterotrophy	Students should be able to: 1. Define the terms autotrophy and heterotrophy.	1. Autotrophy and heterotrophy (i) producers (autotrophs) (ii) consumers (heterotrophs) (iii) aquatic and terrestrial examples of producers and consumers.	Show samples of autotrophs and heterotrophs to students.	Students collect organisms and classify them as producers, consumers and decomposers.	Plants and animals.	State the distinguishing feature between an autotroph and a heterotroph.
II. Food webs and trophic level	Student should be able to: 1. Recognise that food relationship exists among living things. 2. Recognise that chemical energy (in the form of carbohydrates, fats, proteins) and nutrients are transferred among producers, consumers and decomposers. 3. State that trophic level refers to the parts of food chain. 4. Correctly define (or describe) food chains and pyramid of energy/number. 5. Describe the nature of energy transfer or flow in the ecosystem.	1. Trophic levels (a) Energy relationship: Concepts of (i) food chain (ii) food web. (b) Non-cycle nature of chemical energy transfer. (c) Nutrient movement.	Guide students to make chart showing relationship among organisms e.g. food chain, food web.	Students develop a chart showing relationships among organisms (food chain, food webs) Students construct and explain pyramid of energy/number	Cardboard paper, marker. Cardboard paper, marker.	1. Students to differentiate between a food web and a food chain. 2. Using different organisms, produce two examples each of food chain and food web. 3. Students to define pyramid of energy and number.

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TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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	6. State that there is a progressive diminution of energy in the feeding chain. 7. Recognise a definite change in number of individuals from one feeding level to another, especially between producers and consumers.	2. Energy flow (i) Food/energy relationship in aquatic and terrestrial environments. (ii) Pyramid of energy/numbers. (iii) Nature of energy flow as described by food chains.				4. Distinguish between pyramid of energy and pyramid of numbers. 5. Use a food chain to describe energy flow in an ecosystem.

THEME 4: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
10. ENERGY TRANSFORMATION IN NATURE	Students should be able to: 1. Use the knowledge of energy losses in the ecosystem to explain the pyramidal shape of feeding relationships. 2. State that only a small percentage of the radiant energy actually gets to plants. 3. State the first and second laws of thermodynamics and use them to explain ecological events such as pyramid of energy, food chain, energy flow.	1. Energy loss in the ecosystem: (i) Solar radiation: its intake and loss at the earth's surface (ii) Energy loss in the biosphere (iii) measures of primary production e.g. the amount and rate of energy fixation. 2. Laws of thermodynamics: - First law. - Second law. Application of both law to ecological phenomena.	1. Teacher draws on the chalkboard the pyramidal shape of feeding relationships. 2. Teacher uses the law of thermodynamics to explain energy flow across the trophic levels.	1. Students discuss energy loss as a limiting factor in primary production (i.e. production of autotrophy). 2. Students recall instances which are explained by laws of thermodynamics.	Charts and cardboards.	Students to: 1. State why the tip of the pyramid of energy has the least number of organisms. 2. State the first and second laws of thermodynamics.

THEME 4: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
11. RELEVANCE OF BIOLOGY TO AGRICULTURE	Students should be able to: 1. Classify plants using botanical and agricultural techniques of classification. 2. Describe the effects of various agricultural activities on ecological systems. 3. Identify pests of certain crops and indicate their control. 4. Describe some common diseases caused by pests and their control. 5. Identify factors that affect production of crops 6. Identify some methods of preserving and storing foodstuff. 7. List the factors that (i) affects population growth (ii) affects the availability of man's food.	1. Classification of plants: (i) Botanical classification (e.g. algae, spermatophytes) (ii) Agricultural classification (e.g. fibres, latex) (iii) Classification based on life cycles (e.g. annuals, perennials) 2. Effects of agricultural activities on ecological systems (i) Bush clearing/burning (ii) tillage (iii) fertilization/herbicide and their effects (iv) effects of different types of farming on ecological systems 3. Pests and diseases of agricultural importance (i) knowledge of pests (types, life cycles, control) (ii) diseases (types, control)	1. Teacher shows students various types of plants and classifies them 2. Teacher leads students to visit farm projects to observe various effects of fertilizers and herbicides 3. Takes students on field trip to a farm land cleared by burning 4. Takes students to a livestock farm to identify pests of animals	1. Students collect specimens of various plants and classify them using the three techniques. 2. Students discuss the effects of human activities on natural ecosystems. 3. Students note the effects of pests and diseases on plants and animals in the farm. 4. Students make a table showing local farm, livestock pests and diseases the y cause. 5. Students set up experiments to show how food can deteriorate in storage. 6. Students practice different modes of food preservation. 7. Students carry out an experiment to show the effect of limited food on animals population e.g. mice, lizards etc	1. Different types of plant. 2. Charts showing different plant types. 3. Small birds. 4. Diseased plant types. 5. pre-mature fruits shed by diseased plants. 6. different types of plant pests e.g. cassava mosaic.	Students to: 1. Make a large chart showing different ways of classifying plants. 2. Discuss the effects of bush burning on a piece of farm land. 3. List some plant pests and discuss their economic importance. 4. Discuss 4 common ways of preserving farm products. 5. List the consequences of shortage of food (food crisis) on the population.

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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		4. Food production and storage: (i) Ways of improving crop yield. (ii) Causes of wastage. (iii) Methods of preserving and storing food. 5. Population growth and food supply. Reproduction and population growth. 6. Relationship between availability of food and human population. Effects of food storage. 7. Government efforts to increase food production e.g. agricultural revolution.	5. Demonstrate different food storage methods to students e.g. drying process. 6. Collect a large number of insects and use them to demonstrate the effect of food shortage.		1. Fresh vegetables. 2. Dry vegetables. 3. Fresh cassava. 5. Processed garri.	

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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12. MICRO-ORGANISMS AROUND US	Students should be able to: 1. Identify some micro-organisms present in air and water. 2. State the characteristics (colour, pattern of growth, appearance) of colonies of microorganisms. 3. Identify microorganisms that can be found on and inside human bodies e.g. under the fingernail, mouth cavity and expired air. 4. Describe the effects (beneficial and harmful) of microorganisms on human body	1. Micro-organisms in air and water: (i) group of micro-organisms: bacteria, viruses, some algae, protozoa and some fungi (ii) concept of culturing 2. Identification of microorganisms in: (i) air (ii) pond water (iii) river (iv) stream 3. Micro-organisms in our bodies and food: ways in which and places through which micro-organisms enter our bodies 4. Carriers of micro-organisms. Examples of carriers of microorganisms, location of microorganisms in carriers. Types of microorganisms	1. Divide the students into working groups, provides each group with sterilized Petri-dishes which contain culture medium. 2. Coordinate the activities of the students and have them discuss their experiments. 3. Mount some specimen from the different Petri-dishes under a microscope and ask pupils to observe. 4. Put up a chart showing the effects of microorganisms.	1. Students in each group grow cultures of microorganisms from air, water and under the finger nails. 2. They should leave the cultures for three days and then observe, record and make inferences. 3. Performs experiments: (i) to show presence of microorganisms on the teeth, in expired air and dirt under the nails; (ii) to show occurrence of microorganisms on a carrier e.g. housefly. 4. View the micro-organisms under the microscope to identify their pattern of growth. 5. Suggest the effects of microorganisms on food, body, etc.	1. Petri-dishes, culture medium 2. Slides, slides covers, microscopes, stains. 3. Charts showing: (i) some microorganisms and where they are located in humans (ii) Some carriers of microorganisms	Students to: 1. State where micro-organisms can be found. 2. State what could happen to the teeth if not properly brushed for a long time. 3. Describe the characteristics of some microorganisms. 4. What are the effects of micro-organisms on human, animals, etc.

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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(FRESH WATER HABITAT)	Students should be able to: 1. Recognise the variety and size of fresh water habitat. 2. Recognise the variety, quantity and distribution of various organisms in fresh water habitat. 3. Recognise seasonal changes in the size and population of the habitat. 4. Recognise the adaptation of the animal and plants in the chosen fresh water habitat. 5. Estimate the proportion of mineral salt present in the freshwater habitat. 6. Infer the food chain in the chosen freshwater habitat.	1. Characteristics of freshwater habitat. 2. Types of freshwater: (i) Stagnant ones. (ii) Running water. 3. Adaptive features of fresh water organisms. 4. Fresh-water organisms.	1. Take the students to a specified site of study (a fresh water habitat) e.g. pond and guides them to carry out the activities listed. 2. Ensures that students are not taken to infested fresh water e.g. with guinea worm.	On a prepared site, pupils are led by their teacher to: (1) Observe the pattern of distribution of the biotic components of the site and their adaptive features (2) Students identify the biotic components (3) Collect samples from the site (4) construct food chains of the biotic components (5) Take readings of temperature, relative humidity and estimate light intensity (6) Estimate light penetration using sashay disc (7) Estimate movements of water and speed of flow (8) Determine pH, depth, size and turbidity.	1. Sashay disc and other ecological instruments. 2. Charts showing characteristics of freshwater habitat, and plants and animals found in them. 3. Map of Nigeria showing different habitats.	Students to: 1. State the characteristics of freshwater habitat. 2. State some adaptation of animals and plants in freshwater habitat.

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16. TERRESTRIAL HABITAT 1. MARSH	Students should be able to: 1. Recognise types of marshes. 2. Correlate the effect of rainfall or any other source of water and evaporation to the changes in a marsh. 3. Recognise the adaptations of organisms in marshy habitat. 4. Appreciate the marsh as being transitional between aquatic and terrestrial habitat.	1. Characteristics of a marsh. 2. Formation of marshes. 3. Types of marshes. 4. Plants and animals that live in marshes. 5. Adaptive features of these plants and animals.	1. Identify a marsh where to take students for excursion. 2. Guard against students entering a marsh by letting them know the risks involved.	On a pre-selected site, students (1) Observe the pattern of distribution of the biotic components of the site and their adaptive features. (2) Identify the terrestrial animals that usually visit the marsh and other biotic components. (3) Collect samples from the site. (4) Construct food chains of the biotic components. (5) Take readings of soil and air temperature, relative humidity and estimate light intensity.	Ecological implements e.g., shovels, trowels, containers for collecting samples, charts showing marshy areas Map of Nigeria outlining the marshy areas.	Students to: 1. State types of marshes. 2. List the names of five terrestrial animals that visit the marsh. 3. Give three examples of plants that thrive in marshes.

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TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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15. AQUATIC HABITAT (MARINE HABITAT)	Students should be able to: 1. Describe the characteristics of marine habitats. 2. Describe the pattern of distribution of plants and animals in marine habitat, noting the dominant ones. 3. Recognise some adaptive features of the plants and animals in the habitats. 4. Infer the food chain of the organisms. 5. Determine some of the physical factors, e.g. temperature, relative humidity, light, wind and pH.	1. Characteristics of a marine habitat. 2. The major zones: (i) Inter-tidal zone. (ii) Littoral zone. (iii) Oceanic zone. 3. Distribution of the organisms in the habitat. 4. Adaptive features of marine organisms e.g. bladder for floating, holdfast for attachment.	Teacher takes students to study a specified site of (The shallow water near the shore [Littoral Zone] edge of the sea [inter-tidal Zone]).	1. Observe the pattern of the distribution of the biotic components in the habitats and their adaptive features. 2. Identify the biotic components. 3. Collect samples from the site and study them taking note of adaptive features. 4. Construct food chains of the biotic components. 5. Take readings of temperature, relative humidity and estimate light intensity.	Ecological instruments e.g. such discs used to compare turbidity of water, container for collecting samples.	Students to: 1. State the major zones of marine habitat. 2. State common adaptive features found in marine organisms.

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TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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(ESTUARINE HABITAT)	Students should be able to: 1. Describe the characteristics of the estuarine habitat. 2. Describe the pattern of distribution of plants and animals in estuarine habitat, noting the dominant ones. 3. Recognise some adaptive features of the plants and animals in the habitat. 4. Infer the food chain of the organisms. 5. Determine some of the physical factors, e.g. temperature, relative humidity, light, wind and pH.	1. Characteristics of estuarine habitat. 2. Types of estuary. 3. Distribution of the plants and animals in estuarine habitat. 4. Adaptive features of plants and animals in estuarine habitats.	Guide students to study a specified site. Emphasizes the significance of osmo-regulation to estuarine animal.	1. Observe the pattern of distribution of the biotic components of the study and their adaptive features. 2. Identify the biotic components. 3. Collect samples from the site. 4. Construct food chains of the biotic components. 5. Take readings of temperature, relative humidity and estimate light intensity.	Ecological instruments e.g. light meter, containers for collecting samples.	Students to: 1. State three characteristics of estuarine habitat. 2. State some of the physical factors operating in this habitat. 3. What is osmo-regulation, and why is it important for estuarine organisms.

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TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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13. MICRO-ORGANISMS IN ACTION	Students should be able to: 1. Recognise the presence of microbes by the colonies they form. 2. Measure the rate of growth of microbes. 3. List the beneficial effects of microbes from everyday experiences. 4. Recognise that some micro-organisms cause diseases. 5. Recognise that some disease causing microorganisms are air-borne, water-borne and are spread through our food.	1. Growth of micro-organisms: ways of measuring the growth of microorganisms. 2. Beneficial effects e.g. in nature, medicine and industries. 3. Harmful effects of some microbes (i) Types of disease-causing microorganisms (ii) Diseases caused by microorganisms (iii) Ways in which disease causing (Pathogenic) microorganisms spread and are transmitted.	1. Demonstrates culturing and monitor growth rate of microorganisms. 2. Discusses with students the economic importance of microorganisms, and give students projects to carry out to assess the effects of microorganisms.	1. Measure and record growth of a given microorganism from a prepared culture; plot a growth curve and interpret it. 2. Perform experiments on fermentation to produce alcohol, fruit juice, etc. 3. Students produce yoghurt from fresh milk. 4. Students list the names of some microorganisms and the diseases they cause.	Graph sheets. Palm wine, Yeast, sugar, milk, orange, juice, pineapple juice. Charts showing mode of spread of few microorganisms	Students to: 1. State the relationship among growth rate of microorganisms and environmental factors. 2. State the main ingredients for the production of alcohol and yoghurt. 3. List some diseases caused by microorganisms. 4. Discuss some beneficial effects of microorganisms.

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14. TOWARDS BETTER HEALTH	Students to be able to: - Describe some ways by which disease-causing microorganisms and infectious diseases can be controlled. - State ways of controlling vectors. - List ways of protecting ourselves from diseases caused by microorganisms spread by vectors. - Describe some methods used in disposal of refuse and sewage. - State the roles the individuals should play to ensure good health - Name some national and international health organisation and provide some descriptions of what they do.	- Control of harmful microorganisms: Control of disease-causing microorganisms, high temperature, antibiotics, high salinity, antiseptics, dehydration. - Vectors: - definition - ways of controlling vectors. - Students' health: Maintenance of good health - importance to community - ways in which communities do these: - refuse disposal - sewage disposal - protection of water - protection of food - control of diseases - health organizations.	- Teacher organises and takes students to visit sewage treatment plant. - Teacher invites resource persons to talk on public health and get students to write and submit essays on refuse/sewage disposal . - Teacher gets students to construct proper waste disposal sites. - Display charts on - proper waste disposal - refuse treatment - Public health organizations.	- Students work in groups to perform different aspects of the experiment on control of microorganisms, using high temperature, antibiotics etc. Students collect results of the separate experiments and discuss. - Students perform experiment to show that spraying a body of stagnant water with a layer of oil can destroy a vector e.g. mosquito. - Students study the stages in the life cycle of the mosquito. Students record their observations using diagrams and write ups. - Students study charts on health organizations and their areas of work.	Ovens, anti-biotics, Petri-dishes. Charts showing proper refuse treatment and disposal modes. Charts listing health organizations and their areas of work.	Students to: - List some vectors of microorganisms. - How can we protect our environment from harmful microorganisms. - Students to list ways by which water can be protected from pollution. - List 4 national and international health organizations, describing their area of work.

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II. FOREST	Students should be able to: 1. Recognise that water is not always as available to organisms in terrestrial habitats as to those in aquatic habitats. 2. Recognise that water conservation is an important problem in terrestrial habitats. 3. Mention effects of climatic factors on vegetation. 4. Identify the characteristics of the forest habitat. 5. Recognise trophic levels and distribution of animals in a forest. 6. Recognise the stratification of plants in a forest. 7. Construct food chains available in a forest.	1. Characteristics of a forest. 2. Strata in the forest. 3. Distribution of plants and animals that inhabit a forest. 4. Adaptative features of the plants and animals.	1. Identify a forest site to take students on an excursion. 2. Let the students know the inherent dangers in forests and equip them to take precaution.	On a pre-selected site students: (1) Observe the pattern of distribution of the biotic components of the site and their adaptive features. (2) Identify the biotic components. (3) Collect samples from the site. (4) Construct food chains of the biotic components. (5) Take readings of temperature, relative humidity, and estimate light intensity.	1. Ecological materials: hoes, shovels, containers for specimen. 2. Map of Nigeria showing typical forests. 3. Chart showing the layers of a typical forest.	Students to: 1. List some biotic components of the forest. 2. State three characteristics of the forest habitat. 3. Describe plants stratification of the forest habitat. 4. Give examples of plants that are found in different strata of the forest.

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TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
III. GRASSLAND	Students should be able to: 1. Recognise that the problem of water conservation is greater in the grassland than in the forest. 2. Notice the progressive absence of water demanding trees and grasses from the grassland. 3. Recognise the dominant climatic factors. 4. Describe the structural and other adaptations of grassland plants and animals. 5. Identify the predominant plant and animal species and the energy relations between them. 6. Describe the soil structure of grassland.	1. Characteristics of grassland. 2. Types of grassland. 3. Distribution of plants and animals in a grassland. 4. Some adaptations of grassland communities: (i) air-conditioned nests e.g. by termites (ii) thick barks e.g. baobab and palms (iii) leaf fall (iv) underground stems e.g. grasses.	1. Identify grassland where to take students to. 2. Visit the site to ensure its suitability for study. 3. Equip the students to undertake the proposed visit.	On a pre-selected site, students (1) Collect samples. (2) Observe the patterns of distributions. (3) Identify collated biotic components. (4) Take readings of relative humidity, temperature, estimate light intensity. (5) Collect and observe the soil texture and infer the porosity and poor water retentivity.	1. Ecological implements e.g. hoes, shovels, containers for specimens. 2. Map of Nigeria showing the grassland regions. 3. A picture showing a typical grassland or savanna with animals inhabiting it.	Students to: 1. State three dominant diametric factors in a grassland. 2. List names of some predominant plants and animals in a grassland habitat. 3. Identify some characteristics of plants and animals found in grasslands that made them suitable for tier habitat.

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
IV. ARID LANDS	Students should be able to 1. Recognise arid lands as places where water is not available to organisms because it is scarce or frozen. 2. Mention that sun, extremes of temperature, and water scarcity are factors that desert lives have to cope with. 3. Differentiate cold deserts (Tundra) from hot deserts (Sahara). 4. Adaptation of organisms to arid lands.	1. Characteristics of arid lands. 2. Types of arid lands. 3. Distribution of the organisms in the habitat. 4. Some adaptations of organisms to arid lands: (i) water conservation in plants and animals (ii) body temperature regulation in plants and animals.	1. Identify an arid land to take students on excursion. 2. Visit the site to judge its suitability for the study.	On a pre-selected site, students: (1) Observe the pattern of distribution of the biotic components under and/on litter, on trunks, branches and leaves of trees. (2) Collect samples of biotic components. (3) Identify collected biotic components. (4) Take readings of relative humidity and estimate light intensity. (5) Students should take note of soil on vegetations.	1. Ecological instruments that will measure factors in the habitat. 2. Map of Nigeria showing the arid regions. 3. Charts showing characteristics of the arid lands and the components.	Students to: 1. State some characteristics of arid lands. 2. Describe some adaptations exhibited by organisms in the arid habitat.

THEME 4: CONTINUITY OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
17. REPRODUCTION IN UNICELLULAR ORGANISMS AND INVERTEBRATES	Students should be able to: 1. Describe reproduction in: (i) Amoeba. (ii) Paramecium. (iii) Spirogyra. 2. Describe reproduction in: (i) Earthworm. (ii) Cockroach. (iii) Housefly. (iv) Snail. 3. Differentiate between complete and incomplete metamorphosis.	1. Reproduction in Amoeba by asexual reproduction: (i) Binary fission. (ii) Multiple fission. 2. Reproduction in Paramecium by: (i) Asexual reproduction. (ii) Sexual reproduction. 3. Reproduction in Spirogyra by: (i) Asexual or vegetative reproduction. (ii) Sexual reproduction or conjugation. 4. Reproduction in the Earthworm (i) sexual reproduction only. 5. Reproduction in Cockroach (i) Fertilization is internal (ii) Exhibits incomplete metamorphosis.	1. Mount specimens from the ditch on the microscope for students to observe. 2. Request students to collect eggs of cockroaches and observe. 3. Collect maggots of housefly and keep for observations. 4. Collect eggs laid by a snail and keep in a closed container for observation.	1. (a) Collect water from the ditch and view under the microscope to observe unicellular organisms. (b) Observe prepared slides of their reproductive stages. 2. Students to watch earthworm at night with a torch light. 3. Observe the collected cockroach eggs and note your observations. 4. Observe the housefly maggots daily and note your observations. 5. Observe the snail eggs and take daily notes.	1. Prepared microscope slides of conjugation in Spirogyra, microscope slides, torch light. 2. Charts illustrating binary and multiple fission. 3. Chart showing stages of metamorphosis in cockroach. 4. Eggs of cockroach. 5. Maggots from housefly. 6. Snail eggs.	Students to: 1. Define asexual and sexual reproduction. 2. Describe the 2 major ways by which Spirogyra and Paramecium reproduce. 3. Describe the mode of reproduction found in the earthworm. 4. Why is the cockroach said to exhibit incomplete metamorphosis? 5. Distinguish between complete and incomplete metamorphosis.

THEME 4: CONTINUITY OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
		6. Reproduction in housefly: (i) Fertilization is internal. (ii) Exhibits complete metamorphosis. 7. Reproduction in the Snail: (i) Fertilization is internal. (ii) Lay eggs which hatch into young snails.	5. Collect some earthworms and keep in a suitable container for students to observe.	6. Prepare a chart to show complete metamorphosis.	1. Charts showing complete and incomplete metamorphosis. 2. Cardboard sheets. 3. Markers. 4. Containers.	6. What is the major difference in the life cycle of a cockroach and a housefly? 7. What is the main feature of reproduction in snail?

THEME 1: ORGANISATION OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
CLASSIFICATION OF PLANTS	Students should be able to: 1. Classify plants using (a) botanical and (b) agricultural techniques of classification. 2. Classify plants based on life cycles.	1. Botanical classification: (i) Thallophytes (Algae) (ii) Spermatophytes (the seed bearing plants) 2. Agricultural classification: (i) Cereals – maize, rice; (ii) Legumes – cowpea, groundnuts (iii) Root Crops – cassava, yam. (iv) Vegetables – tomato, okra. Onion (v) Fruits – banana, pawpaw (vi) Beverages and drugs – coffee, tobacco. (vii) Spices – ginger, pepper (viii) Oil – coconut, palm fruit (ix) Latex – natural rubber. (x) Fibres – jute, cotton. 3. Plants classified on the span of their life cycles: (i) annuals (ii) Biennials (iii) Perennials	1. Revises Classification in SS 1 using Specimens like spirogyra, mosses, water ferns, maize, orange and hibiscus. 2. Leads students to classify the specimens using agricultural classification. 3. Asks students to select from the specimens and name the classes they belong to based on life cycles.	1. Classify the specimens according to botanical classification. 2. Classify the crops using agricultural classification. 3. Classify the crops into annuals, biennials and perennials.	1. Spirogyra, liver-warts, mosses, maize, orange and hibiscus. 2. Maize, rice, groundnuts, beans, yam, cassava, tomato, okra, banana, pawpaw, coffee, ginger, pepper, coconut, palm fruit. 3. Rice, beans, maize, cabbage, radish, ginger, mango, coconut.	Students to: 1. Record in their notebooks the phyla of the specimens presented. 2. Mention 2 examples of plants (crops) in each classification or category. 3. Write the 2 examples each of Annuals, Bi-annual and Perennials.

THEME 2: THE ORGANISM AT WORK

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
DIGESTIVE SYSTEM	Students should be able to: 1. Describe with appropriate illustrations different types of alimentary tracts. 2. Explain the feeding mechanism of some animals. 3. Explain how some insectivorous plants feed.	1. Types of alimentary tracts 2. Description and functions of parts of the alimentary canal. 3. Modifications of parts to effect their digestive functions. 4. Feeding habits: categories and mechanisms. 5. Modifications in the feeding habits of: (i) filter feeders. (ii) fluid feeders. (iii) insects feeders. (iv) parasitic and saprophytic feeders. 6. Feeding in protozoa, hydra and mammals.	1. With the aid of dissected specimens, models or charts, teach the alimentary canals of planarian, earthworm, grasshopper, birds and rabbits. 2. Use the specimens and charts of the digestive system of housefly, butterfly, cockroach, tapeworm, sheep and dog to illustrate different types of feeding mechanisms. 3. Explains with well drawn charts the differences in the mouth parts of some insects. 4. Shows students specimens of different teeth to illustrate modifications in feeding habits of animals.	1. Observe the learning materials and compare the various alimentary canals. 2. Observe the specimens noting the various modifications found in the animals. 3. Identify the modified mouth parts.	1. Dissected specimens showing the alimentary canals of earthworm, bird and rabbit. 2. Specimens and/or charts of digestive system of housefly, butterfly, cockroach, sheep and dog. 3. Diagrams of mouth parts of some insects. 4. Teeth models of some animals e.g. dog, man.	Students to: 1. (a) Explain the feeding mechanism of the listed animals, housefly, cockroach and dog. (b) Give reasons for the modifications found in listed animals (dog, man, etc). 2. Describe briefly the modifications and mechanisms associated with a named filter feeder and parasitic feeder.

THEME 2: THE ORGANISM AT WORK

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
TRANSPORT SYSTEM	Students should be able to: 1. Define diffusion. 2. Explain the inadequacy of diffusion alone as a transport system for complex organisms. 3. Explain the necessity of a transport system in complex organisms. 4. Identify source of materials and where they are transported to. 5. Discuss the different types of transportation media such as cytoplasm, cell sap, lymph and blood. 6. Describe the mechanisms of transportation in various animals. 7. Compare and contrast various mechanisms of transportation.	1. Definition of diffusion and osmosis and the principles behind them. 2. Need for diffusion. 3. Transport system in large organisms. 4. Need for substances to move over greater distances. 5. Materials for transportation. 6. Structure of arteries, veins, capillaries, vascular bundles. 7. Media of transportation (i) Fluid as medium of transportation - structure and function of blood (ii) Cytoplasm in small organisms (iii) Cell sap or latex in most plants (iv) Closed and open circulation	1. Set up an experiment on rate of diffusion of a coloured liquid. 2. Instructs students to detach the leaves of pawpaw, cassava, Allemanda theveta (milk bush) or rubber required. 3. Explains the structures of artery /vein etc using well drawn diagrams. 4. Mounts the slide of the young petal of Tradescantia plant. 5. Mounts a slide of blood smear and transverse section of an artery and a vein.	1. Observe the experiments and carefully record the time taken by the coloured liquid to diffuse. 2. Detach the leaves of those plants. 3. Observe and make inferences on the materials the plants transport. 4. Note the structures. 5. Examine the slides and observe and draw noting the cytoplasm and granules around the hair cell. 6. Observe and draw the slides of the blood smear, and transverse section of an artery and a vein.	Diagrams showing arteries, veins, capillaries and vascular bundles. Prepared slide of Tradescantia petal. Prepared slide of artery and vein. Charts of cockroach or grasshopper, toad or frog and a named small mammal. Cosine, a young Herbaceous plant, beaker, blade, glass slide and light microscope.	Students to: 1. Write short notes on the structures of arteries and vascular bundles. 2. Make large labeled diagrams of the specimens. 3. What are the differences between the transport system in small animals and large animals. 4. Name two transportation media in large animals.

THEME 2: THE ORGANISM AT WORK

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
	8. Demonstrate experimentally the flow of materials in plant.	8. Mechanisms of transportation: (i) Organisms. (ii) Multicellular organisms. (iii) Higher animals. (iv) Higher plants. 9. Absorption of water and mineral salts.	6. Dissects: (a) cockroach or grasshopper to expose the heart (b) toad, frog or a small mammal to show the heart and some blood vessels. 7. Guide students to set up an experiment showing absorption of water in higher plants.	7. Examine and draw dissected specimens. 8. Locate the transport vessels. Note the differences in the thickness of the walls of the vessels. 8. Observe the experiment and make inferences.		

THEME 2: THE ORGANISM AT WORK

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
RESPIRATORY SYSTEM	Students should be able to: - Describe the different types of respiratory systems. - List the characteristics of a respiratory surface. - Draw and label some of the respiratory organs of some animals. - Explain the various mechanisms of respiration in some animals, e.g. mammals. - Describe the mechanism of exchange of gases through the stomata of plants.	- Types of respiratory systems: - Body surface respiration - Coelomous respiration - Gills - Tracheal system - Lungs - Mechanisms of respiratory system in: - Higher animals. - Lower animals. - Plants. - Mechanisms of gaseous exchange through the stomata of plants.	- Mounts prepared slides showing respiratory organs of earthworm, tilapia, toad, grasshopper and rat. - Displays the gills removed from a dead tilapia for students to observe the various structures. - Discusses respiratory movements in higher and lower animals and in plants. - Leads students to perform experiments showing effects of turgidity on the opening and closing of stomata using a model of stomata.	- Observe the respiratory organs of the various animals. - Observe the gill fakes, gill arch and the gill filaments and draw. - Make notes on the respiratory mechanisms. - Observe and record their observations.	- Prepared slide, - Gills of tilapia, - Chalkboard, relevant diagrams/Charts. - Cellophane tubing to make models of stomata. - Microscope.	- Draw and label the respiratory organs. - State the number of gills in each chamber and how they are attached. - Explain the mechanisms of respiration in named lower and higher animals and plants. - Describe gaseous exchange through the stomata of leaves.

THEME 2: THE ORGANISM AT WORK

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
EXCRETORY SYSTEMS	Students should be able to: 1. Identify and describe different types of excretory systems in plants and animals. 2. Explain the mechanisms of some of the excretory organs and relate structure to functions.	1. Excretory systems: (i) Contractile vacuoles. (ii) Flame cells. (iii) Malpighian tubules. (iv) Kidney. (v) Stomata and lenticels. 2. Excretory mechanism in: (i) Earthworm. (ii) Insects. (iii) Mammals.	1. Discusses the various excretory systems with the students. 2. Asks students to observe: (i) Malpighian tubules in a dissected cockroach. (ii) Excretory system in a dissected rat. (iii) Internal structure of a dissected kidney of a sheep or pig. 3. (i) Removes the epidermis of the lower surface of a dicotyledonous leaf. (ii) Prepare a wet-mount slide and asks students to examine the kidney shaped guard cells and the stomata in the centre of the cells. 4. Displays charts or diagrams of nephron to students, leads them to identify the different parts and relate the identified parts to the functions they perform.	1. Take part in discussion. 2. Observe and draw the malpighian tubules, the excretory system of rat and the internal structure of the kidney of a sheep or pig. 3. Observe the slide, record the observations and make a labelled diagram of the guard cells and the stomata. 4. Relate parts identified to functions.	1. Dissected cockroach showing malpighian tubules, dissected rat and kidney of the sheep. 2. Dicotyledonous leaf, glass slides and microscope. 3. Charts or diagrams of nephron.	Student to: 1. Identify and describe the different excretory systems presented. 2. Draw and label a Mammalian kidney.

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AIDS / TEACHING MATERIALS	EVALUATION CRITERIA
			TEACHER	STUDENTS		
NUTRIENT CYCLING IN NATURE	Students should be able to:	1. Carbon and Oxygen Cycles:	1. With the help of the carbon cycle chart, leads students to	1. List the sources of carbon dioxide.	1. Chart showing the carbon cycle.	Students too
	1. Describe how carbon circulates in nature.	(a) The Carbon Cycle	(a) List the sources of carbon dioxide.	2. Draw the carbon cycle.	2. Beaker, test tubes, green plant (Elodea), filter funnel, water.	1. Describe the process involved in the carbon cycle.
	2. Draw the carbon cycle in detail.	(i) Definition.	(b) Appreciate the relative importance of the various components in the cycle.	3. Perform the photosynthesis experiment.	3. Slides, mirrors or windows louvers, students breathe out on the slides, mirror or louver blade.	2. State two reasons why atmospheric carbon dioxide must be kept in check (i.e. the carbon-oxygen balance).
	3. State why the carbon cycle is necessary for life.	(ii) The process of carbon cycling.	2. Leads students to perform experiment to show absorption of carbon dioxide and release of oxygen during photosynthesis.	4. Observe the experiment and describe the conclusion of water in expired air.	4. Chart showing the nitrogen cycle.	3. Draw and label the detail of the water cycle.
	4. Recognise the delicate balance between carbon and oxygen.	(iii) The importance of carbon cycle for life.	3. Demonstrates the experiment that shows the presence of water in expired air.	5. Draw the water cycle.	5. Decaying materials such as meat broth, fish, boiled rice, conical flask and stopper.	4. Draw the nitrogen cycle.
	5. Describe the part played by plants and animals in the water cycle.	(b) The Oxygen Cycle	4. Discusses the cycle with a well drawn chart.	6. Draw the nitrogen cycle.		5. List the importance of nitrogen to plants and animals.
	6. Draw the water cycle in detail.	(i) The process of oxygen cycling.	5. (i) With the aid of a well drawn chart of the nitrogen cycle, explain the cycle of nitrogen in atmosphere to plants and animals.	7. Observes the results and record their observations.		
	7. Describe With the aid of diagram the role of nitrogen.	(ii) The importance of oxygen to living organisms.	(ii) Importance of nitrogen to plants and animals.			
		(iii) The carbon-oxygen balance				
		2. The Water Cycle:				
		(i) The process of water cycling				
		(ii) Importance of water to living organisms.				
		3. The Nitrogen Cycle:				
		(i) Importance of nitrogen to plants and animals.				
		(ii) The process of nitrogen cycling				

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
	8. State that energy can be obtained by decomposing organic substances. 9. Identify one of the gases produced during decay.	4. Decomposition in Nature: (i) Micro and macro decomposers. (ii) Role of decomposers.	6. Mounts experiments to show that: (a) Gases e.g. carbondioxide, ammonia are released during decomposition. (b) Heat energy is released during decomposition.	8. Identify the gas(es) released.	6. Vacuum flask, boiled beans.	6. State the role of decomposers.

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
ECOLOGICAL MANAGEMENT I. Association	Students should be able to: 1. Recognise some of the different types of association existing between different species. 2. Identify beneficial harmful and neutral forms of association among organisms. 3. Deduce the mode of life of a given organism from observed characteristics.	1. Types of association: (i) Symbiosis - mutual benefits. (ii) Parasitism - harmful to host. (iii) Mutualism - beneficial to both independent living possible. (iv) Commensalism - host unaffected while the commensal benefits. 2. Features of biological importance possessed by organisms of an association.	1. Teacher presents specimens of (i) root nodules (for symbiosis) and (ii) Ascaris (for parasitism) 2. Asks students to collect ticks from cow or dog and where possible hermit crab.	1. Students study the root nodules and the Ascaris noting their characteristic features. 2. Collect ticks from cow or dog.	Root nodules. Ascaris, ticks.	Students to: 1. List the characteristics that make tick an ectoparasite.
II. TOLERANCE	Students should be able to: 1. Discuss why living things possess a range of tolerance to environmental factors. 2. List the abiotic factors that impose tolerance on organisms. 3. Depict tolerance range with a graph.	1. Concepts of minimum and maximum range of tolerance. 2. Geographic range graph illustrating maximum and minimum range of tolerance.	1. Teacher guides students to perform experiment showing limits of tolerance of Tilapia. 2. Teacher draws a typical tolerance range graph.	1. Students subject Tilapia to different concentrations of salt solution and record their observations of rate of opercula movements. 2. Students study the graph.	Tilapia, salt, water, beakers or big glass bowl (depending on fish size), the graph drawn on the chalkboard.	Students to: 1. State three abiotic features for tolerance. 2. Draw a graph of tolerance range.

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
III. ADAPTATION	Students should be able to: 1. State that adaptation may be a modification in response to environmental factors. 2. Describe the availability of water as the principal factor for plant and animal distribution. 3. List examples of adaptation to environmental factors.	1. Adaptation in form and function of living organisms due to environmental conditions. 2. Effects of availability of water on adaptive modifications. 3. (i) Structural adaptation of Tadpoles and fish to life in water. (ii) Structural adaptation in birds.	1. Asks students to examine the root of Hibiscus or any herbaceous plant and the root of a water weed (e.g. Elodea). 2. Places chameleon against different backgrounds. 3. Guides students to attach a lump of plastiline to the base of two water weeds, later drop one into a tall glass jar containing water and leave the other one on the table. 4. Asks student to compare tadpole and fish, fish and toad, sparrow and owl.	1. Relate observed differences to the habitats of the two plants. 2. Note the colour changes. 3. Note and record the difference in the appearance of the two weeds after a while. 4. Note the morphological differences in the listed organisms.	1. Hibiscus plant, water weed, Chameleon, different colours of cloth. 2. Tall glass jar, water, plastiline, two pieces of water weeds Specimens of tadpole, fish, toad, beaks of the owl and sparrow or other birds.	Students to: 1. Explain the principle behind adaptation. 2. Relate the difference in appearance of water weeds. 3. Relate the observed differences in appearance of aquatic and terrestrial adaptations.

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
IV. POLLUTION	Students should be able to: 1. Define pollution. 2. Name air pollutants and their sources. 3. Enumerate the harmful effects of pollutant. 4. State effect of detergents, insecticides, artificial fertilizers and herbicides on aquatic and terrestrial organisms. 5. Identify poor sewage system as a source of pollution. 6. Name domestic and industrial wastes that pollute land and water. 7. Identify the decay of organic matter (e.g. dead animals and plants) as a source of pollution.	1. Pollution of the atmosphere: (i) Nature, names and sources of air pollutants. (ii) Effects of air pollutants. (iii) Effects of noise. 2. Pollution of water and soil: (i) Types and composition of pollutants. (ii) Side effects.	1. (i) Leads students to set up experiments on pollution e.g. set up jars containing water for drinking. (ii) In each jar, place mosquito larvae, tadpoles. (iii) Add pollutants such as crude oil, detergents or insecticide. 2. Leads students to visit an industrial estate to study industrial waste and their effects.	1. Observe the effect of the pollutant on the organisms in the jars 2. Assist to set up experiments on pollution. 3. Observe the various effects of industrial wastes on the environment.	1. Glass jars, mosquito larvae, tadpoles, crude oil, detergents, insecticides. 2. The industrial estate visited.	Students to: 1. Discuss the effect of pollutants on organisms. 2. Write a comprehensive report of their visit in the context of pollution in the environment.

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
CONSERVATION OF NATURAL RESOURCES	Students should be able to: 1. Define conservation. 2. Enumerate reasons for protection of lands and wild life from destruction and extinction. 3. State why various conservation laws should be obeyed. 4. Discuss problems and difficulties associated with conservation. 5. Discuss the need to conserve petroleum and natural gas. 6. State methods of conservation of a given natural resource.	1. Conservation of natural resources: (i) What it means. (ii) Ways of ensuring it: - agencies responsible for conservation. - legislation. - education. - erection of parks and reserves. (iii) Benefits of conservation.	1. Discusses the problems and difficulties of conservation with the students, and lead them to suggest possible solutions. 2. Passes round the conservation law document. 3. Shows film or documentary on gas flaring, oil spillage.	1. Proper solutions to problems of conservation. 2. Browse through the document and study it more closely after class. 3. Watch documentary on gas flaring, oil spillage etc.	1. The conservation law document. 2. The film on gas flaring.	Students to: 1. State the meaning of conservation. 2. State two conservation laws from the law book. 3. Mention two methods of conservation.

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
PESTS AND DISEASES OF CROPS	Students should be able to: 1. Identify pests of certain crops and indicate their control. 2. Describe the life cycles of some of the pests. 3. Describe some 8 common diseases caused by pests. 4. Describe pest control measures.	(A) Pests: 1. Definition of pests. 2. Classification of pest by: (i) The part they attack- (stem borers, root feeders, leaf feeders, young shoot feeders, plant and seed feeders). (ii) Their animal type - (invertebrate pests, e.g. Nematodes, Arthropods, locusts, grasshoppers, coleopteran, Lepidoptera larvae of butterflies). (iii) Vertebrate pests- (birds e.g. quail, weaver bird, mammals e.g. rodents, monkeys). 3. Life cycles of pests. 4. Control of pests. (B) Diseases caused by pest and their agents: (i) fungal diseases (e.g. Blight, root rot and mildew) (ii) viral diseases (e.g. mosaic, molting, stunting) (iii) bacterial diseases (e.g. bacterial wilt, leaf spot, soft rot). (C) Control of pest.	1. (i) Defines pests. (ii) Classify and lead students to give examples. 2. Traces the life cycles of some of the pests. 3. Together with students create a table showing the relationship of crop disease, their agents and methods of control.	1. Give examples of pests. 2. Build the table along with the teacher and copy table in their notes.	Chalkboard, infected leaves and crops. Pests e.g. grasshoppers, termites, locusts, rodents. Pesticides e.g.	Students to: 1. Describe the life cycles of three pests. 2. Describe one crop disease. 3. List three pest control measures.

THEME 4: CONTINUITY OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
REPRODUCTIVE SYSTEMS IN VERTEBRATES	Students should be able to: 1. Identify the various parts of the male and female reproductive systems. 2. Describe the structures and functions of the male and female reproductive organs of vertebrates. 3. Draw and label diagrams of male and female reproductive systems in mammals. 4. Draw, label and describe the structures of male and female gametes (sperm and ovum). 5. Distinguish between male and female reproductive organs in mammals. 6. Distinguish between eggs of vertebrates. 7. Compare reproduction in fish, bird, toad, reptile and mammals.	Reproductive systems in fish, reptile, bird and mammal: 1. Structures of the male and female reproductive systems. 2. Parts of the reproductive systems and their functions. 3. Structures of the male and female gamete (sperm) and the female gamete (ovum). 4. Differences between male and female reproductive organs. 5. Structural differences in the eggs of vertebrates. 6. Comparison of reproduction in fish, reptiles and mammals.	1. Presents to the students dissected male and female vertebrates showing the reproductive organs. 2. Uses the dissected specimens to explain the functions of the organs of the reproductive systems (male and female). 3. (i) Mounts fresh sperm specimen from a rat under the microscope. (ii) Drawing students' attention to the movement. 4. Mounts sections of a mammalian ovary under a microscope.	1. Examine the reproductive organs in the given specimens. 2. Examine the specimens and note the functions. 3. Compare the male reproductive system with the female reproductive system. 4. View the male gamete (rat's sperm) under the microscope. Draw and label the sperm. 5. View the female gamete under the microscope and identify the various parts.	1. Dissected lizard, fish or any other vertebrate showing the male and female reproductive systems. 2. Dissected rats or rabbits showing the male and female reproductive systems. 3. Microscope, slide of the fresh sperm of the rat. 4. Microscope and slide of the ovum. 5. Diagrams of male and female reproductive organs of rat or rabbit. 6. Eggs of toad, fish, lizard, chicken or other available named vertebrates. 7. Chalkboard.	Students to: 1. Draw and label the reproductive system of the male and female mammal. 2. Draw and label the sperm. 3. Tabulate the distinguishing features in sperm and ovum. 4. Discuss three structural differences of the eggs of the various vertebrates. 5. Tabulate the differences in the reproduction of fish, reptiles, birds and mammals.

THEME 4: CONTINUITY OF LIFE						
TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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		7	5. With the aid of diagrams of the male and female reproductive organs of a rat or rabbit, lead students to distinguish between male and female reproductive organs in mammals in tabular form. 6. Displays the eggs of some vertebrates (toad, fish, chicken and lizard) which the students had been asked to bring to class. 7. Leads the students to compare reproduction in fish, toad, reptile, bird and mammals and develop a table showing the comparisons.	6. Distinguish between male and female reproductive organs in mammals. 7. Examine the structural differences in the eggs and record their findings. 8. Compare reproduction in fish, toad, reptiles, birds and mammals.		

THEME 4: CONTINUITY OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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REPRODUCTIVE SYSTEMS IN PLANTS	Students should be able to: 1. Identify reproductive organs in plants. 2. Distinguish between the essential and non-essential parts of a flower. 3. Describe the various reproductive organs of a flower. 4. Describe the different kinds of placentation.	1. Structures and functions of the reproductive organs of plants. 2. Arrangement of reproductive organs in different plants. 3. Types of flowers: (i) hypogynous. (ii) perigynous. 4. Kinds of placentation.	1. Asks students to examine the various parts of a flower provided. 2. Describes each part and points out the essential and non-essential parts. 3. Describes the various reproductive organs of a flower and the position of the ovary in each flower. 4. Prepares transverse section and longitudinal section of ovaries for students to observe with hand lens.	1. Examine the various parts of either Allamanda or Pride of Barbados or Hibiscus and grass. 2. Draw the position of various ovaries. 3. Examine the slides with hand lens and note the different kinds of placentation.	1. Allamanda, Hibiscus, Pride of Barbados, grass 2. Ovaries of Pride of Barbados, Flamboyant flower.	Students to: 1. Identify three various parts of the flower. 2. Draw the ovary position of two flowers.

THEME 4: CONTINUITY OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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POLLINATION IN PLANTS	Students should be able to: 1. Define pollination. 2. Name and describe the different types of pollination. 3. List the features that aid self pollination. 4. State the characteristics of wind pollinated and insect pollinated flowers. 5. State the agents of pollination.	1. Pollination in plants: (i) Types of pollination - self pollination - cross-pollination (ii) Features of self-pollinated flower. (iii) Features of cross-pollinated flowers. (iv) Agents of pollination: - Water. - Insect. - Wind. - Animals.	1. (i) Asks students to study flowers of different kinds and with a diagram: (ii) Relate the position of the stigma and stamens to the type of pollination. (iii) Examine the nature of the pollen grains. 2. (i) Leads students to remove an anther from a flower. (ii) Crush it on a slide, (iii) Prepare a wet mount and examine the pollen grains under the microscope 3. Explains the agents of pollination. 4. Leads students to visit the school garden and observe Insects actively pollinating flowers.	1. Note the positions of the stigma and the stamens. 2. Draw the diagrams. 3. Study the flowers as directed by the teacher. 4. Prepare and examine slides. 5. Discuss observations. 6. Discuss the relationship of stamens and stigma. 7. Write comprehensive report of visit.	Flowers e.g. Hibiscus, Maize Slide, microscope The school garden or any other garden.	Students to: 1. Write two features of self pollinated flowers. 2. List three agents of pollination. 3. Describe two types of pollination.

THEME 2: THE ORGANISM AT WORK

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
REGULATION OF INTERNAL ENVIRONMENT	Students should be able to: 1. List the main organs and substances involved in homeostasis (control mechanism). 2. Describe the structure and functions of these organs: kidney and liver. 3. Name some kidney diseases and explain their symptoms and effects. 4. Name some liver diseases and briefly explain their symptoms and effects. 5. Discuss the remedy for unhealthy conditions kidney and the liver. 6. Explain what hormones are.	1. Homeostatic organs. Substances involved in homeostasis. 2. The kidney: (i) structure, functions and diseases of the kidney: (ii) the effects of kidney diseases and remedy 3. The liver: (i) structure, functions and diseases of the liver, bile products, (ii) the effects of liver diseases and remedy. 4. (i) Endocrine glands (ii) Substances produced by glands (Hormons).	1. Dissect a small mammal to show the liver and the kidney. 2. Shows students photographs or charts of a normal (healthy) and diseased (i) kidney (ii) liver. 3. Shows students charts and photographs that highlights the distinction of endocrine glands. 4. (a) Request students to provide germinated bean seeds. (b) Performs experiments to show the role played by auxins on shoot (phototropism) and root (geotropism).	1. Examine dissected vertebrate e.g. Rabbit and note the following; - structure and position of the kidneys and liver. - relative positions of other organs around them. - number of lobes of the liver. 2. Make a labelled drawing of the liver and organs around it like gall bladder, stomach, pancreas. 3. Observe the distribution of endocrine glands on charts and photographs.	Preserved vertebrate e.g. Rabbit Charts, photographs of endocrine glands Skin of a small mammal on a microscope	Students to: 1. State 3 functions of each of the following organs: (i) Kidney (ii) Liver. 2. Name two kidney diseases of (i) the kidney (ii) the liver. 3 Identify the position of the following glands: - pituitary glands. - adrenal glands in human beings. 4. State the effects of overproduction or underproduction of the following glands: - pituitary.

THEME 2: THE ORGANISM AT WORK

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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	7. List some endocrine glands, their positions in the body and their functions. 8. State the effect of over production or under production of these hormones. 9. Identify the mammalian skin under the microscope. 10. Draw and label main structures of the skin. 11. State the functions of the mammalian skin. 12. State few ways to care for the skin.	(iii) Effects of over production or under production of the hormones. 5. Plant Hormones: - Explain the effects of auxins on lateral bud development, leaf fall and initiations of adventitious roots. 6. Mention other types of plant hormones and functions. 7. Modern application of auxins effects (NAA) with reference to early flowering, crop harvests and weed control (as selective weeds kill dicotyledonous plants). 8. The Skin: (i) Structure of the mammalian skin. (ii) Functions of the mammalian skin. (iii) care of the mammalian skin.	5. Shows students a detailed structure of the mammalian skin under a microscope.	4. Observe and draw the detailed structure of the mammalian skin. 5. Fan themselves and observe sweat on human skin and the cooling effect of evaporation on the skin.	Fan.	- thyroid. - pancreas. - adrenal. - endocrine glands of the gonads. 6. State what size of the shoot have a higher concentration of auxins when shoot is exposed to light from one direction. 7. List five structures found in the mammalian skin. 8. Explain the effect of a fan on someone sweating after a physical exercise.

THEME 2: THE ORGANISM AT WORK

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
NERVOUS CO-ORDINATION	Students should be able to: 1. Locate the position of the CNS (Brain and spinal cord) in a dissected vertebrate. 2. Describe the structure of the brain and explain the functions of its various organs. 3. Describe the spinal cord and explain its functions. 4. State the structural difference between the brain and the spinal cord. 5. Identify the PNS (Peripheral Nervous System) and describe its functions. 6. Describe a typical neuron. 7. Group neurons according to functions as found in vertebrates and explain their functions. 8. Explain the process of impulse transmission.	1. Organisation of the nervous system: (i) CNS (ii) PNS 2. The Brain: Position, structure and functions. 3. The Spinal Cord: Position, structure and functions.	1. Leads students to dissect a small vertebrate to display the brain (in the skull) and the spinal cord. 2. Exposes the brain to show colour of the outer-part. 3. Shows students the permanent slides, or charts of a neuron. 4. Guides students to perform experiments to show reflex action: flicking a handkerchief across student's eyes.	1. Observe the brain and spinal cord in a dissected vertebrate. 2. Observe differences in structure of the brain and the spinal cord. 3. Draw and label the different types of neurons noting the differences in position and structure of basic parts. 4. Explain experience on reflex action.	Photographs/charts of the CNS of a mammal. A dissected vertebrate to show the position of its brain and spinal cord. permanently prepared slides of the transverse section of a vertebrate's CNS. Slides of neurons, microscope, charts or model of a typical neuron. Handkerchief.	Students to: 1. Make a fully labeled diagram of the brain of a dissected rabbit. 2. Name 4 different regions of the mammalian brain. 3. State 2 functions of the spinal cord. 4. State where the spinal cord is located. 5. State the colour of the outer part of the vertebrate's brain.

THEME 2: THE ORGANISM AT WORK

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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	9. Distinguish between a reflex action and a conditioned reflex action giving a few examples of each.	4. PNS (Peripheral nervous system) - Somatic N.S. - Autonomic N.S. - Parasympathetic N.S. - Sympathetic N.S. 5. Structure and functions of a neuron (motor neuron, sensory neuron and relay neuron). 6. Reflex and voluntary actions (Pavlov (1910) experiments on dogs, food and bell).				7. Explain the pathway of impulse when one sits on a bunch of keys placed on a seat without prior knowledge.

THEME 2: THE ORGANISM AT WORK

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
SENSE ORGANS	Students should be able to: 1. List the major sense organs and infer their locations. 2. Describe the organs of smell and taste. 3. Identify the different parts of the tongue associated with sweetness, bitterness, salty taste and sore taste. 4. Draw the skin showing the different receptors present. 5. Describe the structure of the mammalian ear. 6. Explain the functions of the ear – hearing, balance. 7. Describe the mammalian eye and explain the functions of its various parts. 8. Explain the functions of the eye: - image formation - accommodation	1. Organ of smell: (a) The mechanism of smelling with the help of olfactory chemo-receptions. (b) Adaptation for survival – animals that sniff for hunting e.g. Dog - presence of chemoreceptor on butterfly (insects) antennae for sensing its environment. 2. Taste buds on tongue and the part played in tasting. 3. The skin – sensory nerve endings or receptions associated with the skin – touch, deep pressure, - heat and pain. 4. Structure of the ear and part played by its various parts.	1. Leads the students to perceive the releasing of a smelly into the air (e.g. NH_3). 2. Leads students taste bitter leaf, salt and sugar brought from home. 3. Applies various objects on student's skin for sensation of touch. 4. Shows charts/ photographs of an LS of the skin. 5. Show students a photograph, mode or charts of the longitudinal section of the mammalian eye.	1. Smell the released gas. 2. Notes area of tongue, which are sensitive to sweetness, bitterness etc. 3. Notes the different feelings due to the various objects applied on skin. 4. Draw well labeled drawings of the LS of the skin. 5. Study the model, identify its various parts and the pathway of sound waves and vibrations. 6. Use the mirror to observe the papillae of the tongue.	Bottled NH_3 gas. A mirror. charts or photographs showing the different areas of the tongue sensitive to different tastes. Ice cubes/pricking object. A model or chart of the mammalian ear. Charts, photographs and model of the mammalian eye.	Students to: 1. Name the part of the tongue that is sensitive to bitterness. 2. Lists the different sensations felt by the skin. 3. Name the three parts of the ear. 4. Name five parts of the eye and describe their functions. 5. Name the kind of lens used to correct short-sightedness.

THEME 2: THE ORGANISM AT WORK

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
	9. Explain the terms myopia, hyper metropia, astigmatism, cataracts, night blindness. 10. Name the kind of lens that can be used to correct myopic and hyper-metropia defects.	5. Organ of Sight: the eye and its various parts. 6. Functions of the eye and part played by light, image formation and accommodation. 7. Eye defects. : - myopic (short-sightedness) - Hyper-metropic - Astigmatism - cataracts - night blindness	7. With the aid of charts, explains the cause and effect of myopic and hyper-metropic on image formation. 8. Uses charts to explain the correction of myopic and astigmatism with the aid of concave lens, convex lens and cylindrical lens respectively.	6. Identify the parts of the eye. 7. Study the pathway of light during image formation and the placement of lens in front of original lens in order to correct eye defects.	7. Charts on image formation and eye defects. 8. Charts on eye defect correction.	

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
ECOLOGY OF POPULATION	Students should be able to: 1. Define the term succession. 2. Distinguish between primary and secondary succession. 3. Describe the series of changes which often culminate in a stable ecosystem. 4. Discover a definite sequence of plants colonization. 5. Describe the characteristics of a stable community and recognize the climax of succession. 6. Define the term overcrowding.	1. Succession: (i) Meaning of succession. (ii) Structural changes in species composition, variety and increase in number. (iii) Primary succession – in an aquatic habitat, - in a terrestrial habitat. (iii) Secondary succession – meaning and examples. (v) Characteristics of a stable community. 2. Overcrowding: (i) Population density studies vis-à-vis available resources. (ii) Importance of factors affecting population e.g. space. 3. Explain the relationship between competition and succession. 4. Factors that may cause overcrowding e.g. increased mortality, reduced death rate, absence of predators etc.	1. Map out an area on a bare ground and another on lawn that has been cleared earlier on (within the school premises). 2. Take students to a nearby pond. 3. Lead students to measure the size of their class and determine available space for movement of each member of the class when: (a) only the class members are present (b) members of another class are added. 4. Out-door practicals: visit to part of school or close by area with Delonix regia (Flame of the Forest) to see explosive mechanism that aids the dispersal of seeds.	1. Study succession on both pieces of land and record findings over time (about 2 weeks). 2. Compare available space of movement of each member of the class when (a) only the class members are present and (b) members of another class are added. 3. Observe explosive mechanism and note the distance between parent plant and dispersed seeds. 4. Observe the behaviour or activity of the grasshoppers in the two boxes. 5. Observe the behaviour of five students sharing a small seat.	1. Mapped out bare ground and a mapped out piece of lawn that had earlier on been identified. 2. Pond, or lakes to observe colonizers. 3. Students of the class being taught and another class put together. 4. Observing dispersal of seeds from pods of Delonix regia.	Students to: 1. Observe and record different organisms and their member in a lawn over the period of 2 weeks. 2. What was the effect of other students coming to join the members of your class on your activities. 3. Name one animal that migrates during unfavourable conditions. 4. Describe the activity of grasshoppers in each of the boxes presented to you in class.

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
	7. State factors that may cause overcrowding. 8. State the effects of overcrowding. 9. State measures adopted in nature to avoid overcrowding. 10. State the causes of food shortage. 11. List the effects of food shortage. 12. Infer that food has direct influence on mortality on animal population.	5. Discuss measures adopted by nature to avoid overcrowding e.g. dispersal of seeds and fruits, migration of organisms during unfavorable periods, territorial behaviour in some organisms e.g. lizards. 6. Discuss the effects of food shortage: - Competition. - Reproduction. - Emigration. - Increase in mortality rate.	5. Put grasshopper in 2 boxes, one set should be well fed while the other is underfed. 6. Get five students to sit on a desk meant for three students. 7. Display a chart showing agents and means of dispersal. 8. Ask students to describe what they think will happen in a poultry where the feed for the chicks was reduced from 10 to 2 bags a day.	6. Write an essay on the effects of reducing the feed supplied to a poultry from 10 to 2 bags a day.	5. One set of well fed grasshopper in a perforated box and another set of underfed grasshopper in a perforated box. 6. Charts showing agents/means of dispersal.	5. List the factors affecting population. 6. What is the relationship between competition and succession of species. 7. What are the measures taken by organisms in nature to avoid overcrowding. 8. What are the consequences of food shortage on plant and animal population.

THEME 3: THE ORGANISM AND ITS ENVIRONMENT

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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BALANCE IN NATURE	Students should be able to: 1. Recognise population as an integral part of the community. 2. Explain that different population of organisms make up a community. 3. List factors that may effect changes in population. 4. Explain the term dynamic equilibrium as related to different population in a climax community. 5. Recognise the approximately constant numbers of individuals in a population despite fluctuation. 6. List some family planning methods in human community.	1 Factors affecting a population: (i) Abiotic - light, space, heat, water etc. (ii) Biotic - food, competition (inter specific and intra specific), mortality, parasites, predators etc. 2. Dynamic equilibrium in nature and factors that maintain it. (i) Density independent factors like availability of food. (ii) Density dependent factors e.g. natural disasters. 3. Family planning: natural and artificial method used by man to control his population.	1. Map out an area, lawn within the school compound for study. 2. Plant, seedlings to show inter-specific and intra-specific competition. 3. Discusses the process by which omnivores maintain constant population by density dependent and density independent factors. 4. Display a chart showing family planning methods.	1. Note the different population of organisms that inhabit a given lawn. 2. Record observations over time (about 2 weeks) note the number of seedlings that survive after 2 weeks. 3. General discussion on natural disasters e.g. Tsunami, earthquakes, etc. 4. Observe the chart and discuss family planning methods.	1. A lawn around the school premises for study by students. 2. (i) Seedlings of maize planted in a small container e.g. empty tin of evaporated milk. (ii) Seedlings of cowpea planted in a container. 3. Charts showing family planning methods.	Students to: 1. Name three different population on the lawn that make up that community of organism. 2. Explain the terms: (i) Intra specific competition. (ii) Inter specific competition. 3. Are natural disasters avoidable? What are the effects of disasters on population of organisms in an affected area.

THEME 4: CONTINUITY OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
REPRODUCTIVE SYSTEMS AND REPRODUCTION IN HUMANS	Students should be able to: 1. Identify the various parts of the male and female reproductive system in man. 2. Draw, label and describe the structures of the male and female reproductive organ in man. 3. State the function of the various parts of the male and female reproductive systems in man. Compare the structures. 4. Draw, label and describe the structure of the male and female gametes (sperm and ovum). 5. Explain the process of fertilization in man. 6. List the conditions necessary for survival.	1. Reproductive systems in male and female (human): (i) Functions of its various parts. (ii) The structure of a male sperm cell and structure of an ovum. 2. Fertilization: (i) Fusion of gametes. (ii) Implantation. 3. Development of embryo: (i) Selective exchange between mother and child. (ii) Removal of excretory products from foetus. (iii) Survival conditions like food, oxygen, amniotic fluid, placenta and umbilical cord.	1. Shows charts of male and female reproductive systems. 2. Shows charts of male (sperm) and female gametes. 3. Shows charts of fusion (i.e. a sperm cell and an ovum). 4. Shows photographs or charts of a developing foetus.	1. Identify the different reproductive organs in the system. 2. Compare the male reproductive systems with the female. 3. Compare the gametes and note features. 4. Observe and identify the different conditions that aid survival of the foetus.	1. Well labelled charts of a male and female reproductive systems. 2. Charts of a well labelled male and female gametes (sperm and ovum). 3. Charts and photographs showing the developing foetus, its surrounding amniotic fluid, placenta and attachment to the mother's uterus.	Students to: 1. State the differences between the male and female reproductive systems. 2. State what is common to both gametes. 3. State two functions of the umbilical cords. 4. Draw and label: (i) The male reproductive system. (ii) The female reproductive system.

THEME 4: CONTINUITY OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
DEVELOPMENT OF NEW SEEDS	Students should be able to: 1. Define fertilization. 2. Draw and label the male and female reproductive parts involved in fertilization. 3. Describe the process of pollen development. 4. Locate the position of embryo within the seed.	1. Progress of development of zygote in the flowering plant: (i) pistil (ii) stem (iii) germplasm (iv) fertilization. 2. Germination of seeds: (i) Types of germination. (ii) Conditions necessary for germination.	1. Provides students with dissected flowers to show the parts involved in fertilization. 2. Demonstrates experiment to grow pollen. 3. Displays charts showing the growth and movement of pollen. 4. Displays slides showing the various stages of embryo development in flowering plants. 5. Guides students to soak seeds in water and plant some in milk tins to observe types of germination and conditions necessary for germination.	1. Observe the dissected flower noting the male and female parts. 2. Observe the slides and charts showing the growth of pollen and stages of embryo development. 3. Soak seeds in water for different number of days and observe with hand lens. 4. Plant seeds in different containers and uproot to observe on different days. 5. Place the planted seedlings under different conditions to identify the conditions necessary for germination.	1. Prepared slides of pollen growth. 2. Charts on seed development. 3. Empty milk tins. 4. Beakers with water. 5. Assorted seeds – beans, maize etc. 6. Sand. 7. Hand lens.	Students to: 1. Define fertilization. 2. Discuss the role played by the pistil and stamen in fertilization. 3. State 2 main types of germination. 4. List 4 main conditions necessary for germination.

THEME 4: CONTINUITY OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
FRUITS	Students should be able to: 1. Distinguish between fruit and a seed. 2. State various types of fruits. 3. Classify fruits. 4. State the ways fruits are dispersed. 5. Describe the features that adapt fruits to their modes of dispersal.	1. Structure of fruits. 2. Types of fruits. 3. Dispersal of fruits and seeds. 4. Agents of dispersal.	1. Brings various kinds of fruits to the class. 2. Works with students to sort and classify the fruits. 3. Guides students to link physical features of fruits with their mode of dispersal.	1. Brings various kinds of fruits to the class based on teachers instruction. 2. Sort the fruits into various types. 3. Sort out fruits into their modes of dispersal.	1. Fruits. 2. Charts/diagrams of fruits of various types. 3. Chart showing agents of fruit dispersal.	Students to: 1. State the main distinguishing features between a seed and a fruit. 2. Using various categories to classify fruits into various types. 3. Name any three fruits and use their physical features to describe their modes of dispersal.

THEME 4: CONTINUITY OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
REPRODUCTIVE BEHAVIOURS	Students should be able to: 1. State the different types of courtship – behaviours in different animals. 2. Explain the terms: (i) courtship (ii) territorialism in animals.	Courtship behaviours in animals: 1. Pairing 2. Territoriality 3. Display of colours and body parts 4. Seasonal migration	1. Shows students pictures of colourful birds and preserved specimen of Agama lizard. 2. Takes students to where lizards bask to note the courtship behaviour.	Observe and discuss animals e.g. lizards, Cock, turkey exhibiting courtship behaviours.	Pictures of birds, Life animals such as peacock, Agama Lizard, turkey in their natural habitat.	Students to: 1. State three courtship behaviours in animals. 2. Describe the courtship behaviour exhibited by a named male animal.
BIOLOGY OF HEREDITY (GENETICS)	Students should be able to: 1. Identify the dominant/recessive characteristics. 2. Infer that dominant characters mask the contribution of the recessive characters for the first generation (F₁). 3. Recognise that the dominant characters become manifest in more individual members of a population than the recessive characters.	1. Transmission and expression of characters in organisms: (i) Hereditary variations (ii) Characters that can be transmitted and how they are transmitted (iii) How characters manifest from generation to generation. 2. Chromosomes, the basis of heredity (i) Location (ii) structure (iii) role in and processes of transmission of hereditary characters from parents to offspring. 3. Probability in genetics.	1. Discusses with students Mendel's experiment with red and white flowered peas. 2. Uses a projector to show a slide of chromosomes. Charts can also be used to show chromosomes. 3. Shows students the local variety of cassava (a biennial crop) and the Improved cassava (an annual crop).	1. Experiment on cross breeding using <i>Drosophila</i> and maize. 2. Compare your results (i.e. the ratios they obtain) with that of Mendel. 3. Observe permanent prepared slides of root tips of onion or lily showing mitosis. 4. Observe prepared slides of testes of locust or grasshopper.	1. Pictures of birds, animals such as peacock, Agama Lizard. 2. Local variety of cassava (a biennial). 3. Improved variety of cassava (an annual). 4. Charts to show cross between different genotypes e.g. AA x aa.	Students to: 1. Draw a chart of TT x tt characters. 2. State what recessive and dominant characters are. 3. State what chromosomes are. 4. What is probability in genetics? 4. Which of the two varieties of cassava do you think farmers will prefer and why?

THEME 4: CONTINUITY OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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	4. Identify chromosomes in permanently prepared slides of cells. 5. Note that chromosomes carry genes which are responsible for inherited characters. 6. (a) Explain the advantages and disadvantages of: - Cross fertilization and self-fertilization. - Sexual reproduction and asexual reproduction. - Out and in-breeding. (b) Relate the application of the above to practices in agriculture and medicine.	4. Application of the principles of heredity: (i) In agriculture: - Improved varieties. - Diseases resistant varieties. (ii) In medicine advice for couples in relation to the sickle cell gene. 5. Explain the terms: - Cross fertilization. - Self fertilization. - Out and in-breeding using Mendelian crosses.	4. Asks students to do some cross AA x SS (sickle cell gene) 5. Shows students a chart depicting products of cross-fertilisation and self-fertilisation.	5. Discuss the advantages of the improved variety of cassava over the local. 6. Carry out Mendalian Cross to explain offsprings' genotypes.	5. Prepared slides of root tips of onions, testes of grasshopper. 1. Charts showing crossing of AA x SS. 2. Charts showing products of self and cross-fertilisation.	5. Advise someone with the AS genotype not to marry an AS. Show this with a Mendalian Cross. 6. Discuss the disadvantages of self-fertilisation.

THEME 4: CONTINUITY OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
VARIATION AND EVOLUTION	Students should be able to: 1. Define the terms evolution and variation in relation to living organisms. 2. Identify the relationship between the two terms variation and evolution. 3. Recognise that variation could be morphological or physiological. 4. State a few examples of: (i) morphological variations. (ii) physiological variations. 5. List different ways that individuals vary in their behaviours and functions. 6. Explain the application of variation to crime detection. 7. State the blood group types that can be transfused for individual with known blood group. 8. Deduce the possibility of children with particular blood group belonging to a man (father).	Variation and Evolution (1) Morphological variation (physical appearance) which can be: (i) Continuous (size, weight, height). (ii) Discontinuous (a) (colour, skin, eye hair, coat of animal). (b) Finger prints. (2) Physiological variations (i) Behaviour (ii) Ability to roll the tongue (iii) Ability to taste phenylthiocarbamide (PTC) (iv) Differences in blood groups (A, B, AB and O). Application of Variation (i) Crime detection (finger prints) of dangerous weapons. (ii) Blood transfusion. (iii) Determination of paternity.	1. Measures height and weight of students of the same age group e.g. 12 years – 15 years, 14 years to 16 years. 2. Shows photographs of maize cobs, apple and ripe pineapples. 3. Asks students to roll their tongues. 4. Asks students if they know their blood group. 5. Demonstrates the making of fingerprints and identify the different patterns, made e.g. wells, arches etc. 6. Makes a table with different blood groups.	1. Record different heights and weights of students in the same group. 2. Observe and record colour patterns on plants. 3. Attempt to roll the tongue. 4. Name their different blood groups. 5. Students make finger prints, and notes the different types of finger prints.	1. Bathroom weighing scale for weighing students. 2. Materials and markers for measuring heights of students. 3. Students are made to roll the tongue in order to appreciate that this feature is physiological i.e. not everyone can roll the tongue. 4. Stamp pad with ink for thumb printing. 5. A table for blood group match and an empty space to indicate the blood that can be given to individual with particular blood group types.	Students to: 1. Draw a frequency distribution table showing the measurements made and explain the implication. 2. Can everyone roll the tongue. 3. Compare the finger prints made by students. 4. Are there two students with the same pattern? 5. Can a man with blood group AB possibly be the father of a boy whose blood group is B given that the mother's blood is group O?

THEME 4: CONTINUITY OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
			TEACHER	STUDENTS		
EVOLUTION	Students should be able to: 1. Discuss the progressive change in anatomy of organisms from water to land. 2. Explain the term Adaptive colouration and their functions. 3. State examples of structural adaptations in the following organisms that aid their survival: (i) mosquitoes feeding (ii) securing of mates in agama lizards. 4. Explain the term behavioural adaptation as regards the castes in termites. 5. State the law of use and disuse according to Jean Baptist de Lamar.	1. The progressive change in structures and anatomy of organisms using examples of features of organism from water to land. 2. Structural adaptation: Adaptive colouration and their functions (e.g. camouflage in chameleons, Green snakes in green grasses, Bright colours of flowers to attract pollinations etc). 3. Structural adaptation (i) for obtaining food e.g. proboscis for sucking sap in insects. (ii) Protection and defense e.g. tortoise in shell, (iii) Mimicry colours for securing mates for regulating body temperature etc. 4. Different castes of termites and their roles. 5. Different castes of bees and the roles they play in their social life. Theories of Evolution	1. Shows life specimens of tadpoles in the external gill stage, fish with internal gills and toads with lungs. 2. Visits to ornamental plants around the school. 3. Organises visit to the zoo to see a peacock and a peahen. 4. Provides male Agama lizard to show colour. 5. Shows charts and real termites: soldiers, king, queen and reproductive males and females. 6. Discusses with students the reality of the law of use and disuse by referring to boxers or athletes.	1. Observe the progressive changes of respiratory surfaces in animals (structures). 2. Take note of the destination of butterflies. 3. Note the colourful feathers of the peacock against the dull colour of the peahen. 4. Note the colourful bands on the Agama in contrast with the dull green colour of the female lizard. 5. Draw the different castes of termites.	1. Tadpoles in the external gill stage, fish and toads. 2. Butterflies and ornamental plants. 3. Visit to the zoo. 4. Termites. 5. Charts of the different castes of termites. 6. Pictures of boxers, weightlifters and footballers legs.	Students to: 1. State the importance of the association between butterflies and flowers/plants. 2. Give an example of a plant that practices mimicry. 3. Name the two kinds of soldiers in the termite castes. 4. Why do you think Charles Darwin's theory is more popularly accepted? 5. State the structural adaptation in an organism and its function in the life of the organism.

THEME 4: CONTINUITY OF LIFE

TOPIC	PERFORMANCE OBJECTIVES	CONTENT	ACTIVITIES		TEACHING AND LEARNING MATERIALS	EVALUATION GUIDE
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	6. State 2 important features that support the evidence of evolution. 7. State other occurrences apart from natural selection that may bring about evolution.	6. The theories of evolution according to Charles Darwin, Jean Baptist de Lamar - law of use and disuse - law of inheritance or acquired characteristics (related to the work of August Weismann using his experiment of mating of mice). 7. Evidence of evolution: (i) Fossil records. (ii) Embryology. 8. Forces responsible for evolution: mutation, gene flow, genetic drift.	7. Discusses the possibility of transfer of such acquired characteristics e.g. can a footballer's child have big and developed legs even without trainings. 8. Shows students real specimen of <i>Amaranthus spinosa</i> and <i>Amaranthus hybridus</i>. 9. Shows photographs of evolutionary trend in the <i>Homo sapiens</i> e.g. <i>Homo erectus</i>, etc. 10. Provides photographs of the different embryos of vertebrates.	6. Note the efficiency and pronouncement made when a part of the body is in constant use and discuss freely the possibility of such traits being transferred. 7. Study both plants and note the similarities and differences between them. Suggest the reason for the differences from one to the other. 8. Study the similarities between the skeletons of chimpanzee, gorilla and man. 9. Take note of the tail in man during gestation, and common gill stage at one point or the other in the life of an embryo.	6. The vegetable: (i) Edible (<i>Amaranthus hybridus</i>). (ii) the weed (<i>Amaranthus spinosa</i>). 7. Fish, toads and lizards. 8. Photographs showing different stages of evolution in man. 9. Photographs of different embryos of vertebrates.	6. Draw the features that support evolution and indicate common ancestry of organisms. 7. Apart from natural selection, what other forces are responsible for evolution?

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